Bayesian Statistics

Script for the course in autumn 2025

(subject to modifications during the semester)

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Chapter 1

Introduction to Bayesian statistics

In this chapter, we introduce the basics of Bayesian statistics and compare how the Bayesian approach relates to the other main paradigm of statistical inference: frequentist statistics.

1.1 Bayes' formula

In the discrete case, we have a finite or countable partition (A_i) of Ω and another event $B \subset \Omega$ with $P(B) > 0$ (we tacitly assume that all sets we encounter are measurable). Then the following Bayes' formula is well known and easy to prove

$$\begin{aligned} P(A_i \mid B) &= \frac{P(B \mid A_i)P(A_i)}{P(B)} \\ &= \frac{P(B \mid A_i)P(A_i)}{\sum_{k:P(A_k)>0} P(B \mid A_k)P(A_k)} \end{aligned}$$

This formula has both a frequentist and a Bayesian interpretation. In the frequentist interpretation it gives the relative frequency of the event A_i among those repetitions where B occurs. In the subjective interpretation it tells how to modify the prior degree of belief $P(A_i)$ in A_i after observing that B has occurred.

A somewhat counterintuitive implication of Bayes' formula is the so-called base rate paradox: $P(B \mid A_i) \gg P(B \mid A_k)$ does not imply that $P(A_i \mid B) > P(A_k \mid B)$. It only holds that

$$\frac{P(A_i \mid B)}{P(A_k \mid B)} = \frac{P(B \mid A_i)}{P(B \mid A_k)} \frac{P(A_i)}{P(A_k)}$$

and the second factor on the right also influences the value on left. As an example, in case of a rare disease a positive outcome of a test with small error probabilities still can give a low probability of actually having the disease.

Various generalizations of Bayes' formula for continuous situations exist. If (X_1, X_2) is a bivariate random vector with joint density f_{X_1,X_2}, then we have the marginal densities

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1, x_2)dx_2, \quad f_{X_2}(x_2) = \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1, x_2)dx_1$$

and the conditional densities

$$f_{X_2|X_1}(x_2|x_1) = \frac{f_{X_1,X_2}(x_1, x_2)}{f_{X_1}(x_1)}, \quad f_{X_1|X_2}(x_1|x_2) = \frac{f_{X_1,X_2}(x_1, x_2)}{f_{X_2}(x_2)}$$

provided the denominator is neither zero nor infinity, and arbitrary otherwise. Note that we are conditioning on an event with probability zero. Then it follows that

$$f_{X_1|X_2}(x_1|x_2) = \frac{f_{X_2|X_1}(x_2|x_1)f_{X_1}(x_1)}{\int f_{X_2|X_1}(x_2|x_1')f_{X_1}(x_1')dx_1'}.$$

The densities above need not be with respect to the Lebesgue measure, any product measure can be used. Moreover, the existence of densities for the marginal P_{X_1} of X_1 is not needed: If the conditional distribution of X_2 given $X_1 = x_1$ has a density $f_{X_2|X_1}(x_2|x_1)$, then the conditional distribution of X_1 given $X_2 = x_2$ is absolutely continuous with respect to the marginal P_{X_1}, and the density is given by

$$\frac{f_{X_2|X_1}(x_2|x_1)}{\int f_{X_2|X_1}(x_2|x_1')P_{X_1}(dx_1')}$$

for those x_2 where the denominator is neither zero nor infinity and arbitrary otherwise.

1.2 Basics of Bayesian statistics and comparison to frequentist statistics

There are two major paradigms for making statistical inference: the frequentist and the Bayesian one. In the following, we introduce the approach of Bayesian inference and compare the two main statistical paradigms.

Both frequentist and Bayesian statistics assume that the observations x are realizations of a random variable X and aim to draw conclusions about the underlying distribution based on the observed values. We assume that the set of distributions $\mathcal{P}$ which are considered as possible are parametrized by θ:

$$\mathcal{P} = (P_\theta; \theta \in \Theta).$$

In parametric statistics, Θ is a subset of $\mathbb{R}^p$ and thus finite dimensional. In this course, we are mostly focusing on the parametric case. Further, we denote the space where the observations x are by $\mathbf{X}$:

$$X \in \mathbf{X}.$$

Usually, $\mathbf{X} = \mathbb{R}^n$ for some n. The goal of statistical analysis, both frequentist and Bayesian, is to make inference on the parameter θ by using the observed data x.

1.2.1 The Bayesian approach to statistics

Whereas frequentist statistics considers θ to be unknown, but fixed, Bayesian statistics treats θ as a random variable. Because repeated experiments with varying θ's drawn at random usually do not make sense, probability statements about θ have an epistemic meaning in the sense of uncertainty quantification about the parameter before and after seeing the data. The uncertainty about θ before seeing the data is described by a prior distribution π on Θ. The distribution P_θ is then the conditional distribution of X given θ, and we write the density of P_θ as $f(x|\theta)$. This density $f(x|\theta)$ is called the likelihood. A Bayesian model consists of these two parts: a likelihood $f(x|\theta)$ and a prior $\pi(\theta)$. It follows that the joint density $\pi(x, \theta)$ of (X, θ) is given by $\pi(x, \theta) = \pi(\theta)f(x \mid \theta)$.

Bayesian inference is done by conditioning on the observed data x. The posterior distribution is defined as the distribution of θ given $X = x$.[1] The posterior describes the uncertainty about θ after seeing the data. By Bayes' formula the posterior density is given by

$$\pi(\theta \mid x) = \frac{\pi(\theta) f(x \mid \theta)}{\int_\Theta \pi(\theta') f(x \mid \theta') d\theta'}.$$

In general, Bayesian analysis involves the following three main challenges: (i) finding a good model (both prior and likelihood), (ii) calculating the posterior, and (iii) assessing the fit of the model. In this course, we will mostly focus on the first two points (see Sections 2 and 3 for (i) and Section 4 for (ii)). Concerning (i), we note that since the topic of specifying "data models", i.e., likelihoods, is already covered in many other statistical courses, the focus lies on specifying prior distributions in this course. The choice of the prior can be difficult and it can influence the result of the analysis, see e.g. the base rate problem or Example 1.1 below of a normal mean where the posterior can be any normal distribution if we choose the prior mean and variance accordingly. However, one is not allowed to let the prior depend on the data, it has to be chosen before seeing the data. The required choice of the prior is sometimes considered a weak point of Bayesian statistics and it has lead to much controversy. We will discuss it in more detail in Chapter 2, including attempts to define non-informative priors.

Once a Bayesian model is chosen (both prior and likelihood) and one conditions on the observed data, the posterior is completely specified. Depending on the application, calculating the posterior might still be difficult from a computational point of view, and one might want to summarize the posterior in a point estimate depending on the loss function that one uses (see Section 1.2.2). However, how to do inference is essentially determined once a model is specified and the data is observed. A frequentist, on the other hand, starts with any inference method (e.g., a likelihood-based method, any other optimization method, method of moments, etc.) and asks himself what would happen if one repeated the procedure many times with the data changing each time, i.e., the data being sampled again from the same model. A frequentist then compares the different estimators for the parameter and uses the procedure (often maximum likelihood) which has good properties such as consistency, a good convergence rate, low (asymptotic) variance etc.

In the following, we introduce a first example of a Bayesian model.

Example 1.1 (Normal means)**.** We discuss the case where we have n i.i.d. observations from a Gaussian distribution with unknown mean θ and known variance $\sigma^2 = 1$. If we choose $\mathcal{N}(\mu, \tau^2)$ as prior for θ, the density of the posterior is

$$\begin{aligned}\pi(\theta|x_1, \dots, x_n) &= \frac{\prod_{i=1}^n f(x_i \mid \theta)\pi(\theta)}{\int \prod_{i=1}^n f(x_i \mid \theta')\pi(\theta')d\theta'} \\ &= \frac{\exp\left(-\frac{1}{2\tau^2}(\theta-\mu)^2 - \frac{1}{2}\sum_{i=1}^n (x_i-\theta)^2\right)}{\int \exp\left(-\frac{1}{2\tau^2}(\theta'-\mu)^2 - \frac{1}{2}\sum_{i=1}^n (x_i-\theta')^2\right) d\theta'}.\end{aligned}$$

By completing the square in the exponent, the numerator is seen to be equal to

$$\exp\left(-\frac{n+\tau^{-2}}{2}\left(\theta - \frac{\mu + n\tau^2\bar{x}}{1+n\tau^2}\right)^2 - \frac{1}{2}\sum_{i=1}^n (x_i-\mu)^2 + \frac{n^2\tau^2}{2(1+n\tau^2)}(\bar{x}-\mu)^2\right)$$

[1] By a slight abuse of notation, we denote both the distribution and its density by $\pi(\theta|x)$ in this script.

where $\bar{x}$ is the arithmetic mean $n^{-1}\sum x_i$. The same algebraic manipulation can be made in the integrand of the denominator. Because the second and third term in the exponent do not contain θ, they cancel and we are left with a Gaussian integral in the denominator. We therefore obtain

$$\pi(\theta|x_1,\ldots,x_n) = \frac{\sqrt{n+\tau^{-2}}}{\sqrt{2\pi}} \exp\left(-\frac{n+\tau^{-2}}{2}\left(\theta - \frac{\mu+n\tau^2\bar{x}}{1+n\tau^2}\right)^2\right),$$

or in other words

$$\theta \mid (X_1 = x_1,\ldots,X_n = x_n) \sim \mathcal{N}\left(\frac{1}{1+n\tau^2}\mu + \frac{n\tau^2}{1+n\tau^2}\bar{x}, \frac{\tau^2}{1+n\tau^2}\right).$$

The posterior mean is a convex combination of the prior mean μ and the maximum likelihood estimate $\bar{x}$ and the posterior variance is the prior variance divided by $1+n\tau^2$.

The above computations could have been simplified by observing that in the numerator we can ignore any factor which does not contain θ because the same factor will appear in the denominator and thus cancels. Also the denominator is simply a normalization which ensures that the posterior density integrates to one. To emphasize this, one usually writes Bayes' formula as

$$\pi(\theta \mid x) \propto \pi(\theta) f(x \mid \theta)$$

where the proportionality sign means up to a factor which does not contain θ, but may contain x. I.e., Bayes' formula says that "the posterior is proportional to prior times likelihood".

The denominator $f(x) = \int f(x \mid \theta)\pi(\theta)d\theta$ in Bayes' formula for the posterior is called the marginal density, marginal likelihood, or also the prior predictive density of X. Interestingly, the marginal likelihood $f(x)$ contains a form of built-in model complexity penalty and thus suffers less from overfitting compared to the frequentist likelihood $f(x \mid \theta)$. In other words, the marginal likelihood will typically increase with increasing model complexity up to a certain point and, subsequently, it will decrease again with increasing model complexity. Intuitively, if a model is complex, $p(\theta)$ is defined on a large space and will thus typically spread out its mass very thinly both along "good" and "bad" parameters, i.e., parameters for which the likelihood $f(x \mid \theta)$ is large and small, respectively. Since not a lot of mass is put on θ's for which $f(x \mid \theta)$ is large, the integral $\int f(x \mid \theta)\pi(\theta)d\theta$ will thus generally be small. This can also be seen by noting that

$$\log f(x) = \log f(x|\theta) + \log \pi(\theta) - \log \pi(\theta|x)$$

since $\pi(\theta|x) = f(x|\theta)\pi(\theta)/f(x)$. This holds for any θ and, in particular, also for the maximum likelihood estimate $\hat{\theta} = \arg\max_\theta f(x|\theta)$. The first term $\log f(x|\hat{\theta})$ will increase with increasing model complexity. However, the remaining term $\log \pi(\hat{\theta}) - \log \pi(\hat{\theta}|x)$ can be seen as a penalty term whose sign is generally negative since $\log \pi(\hat{\theta}|x)$ is typically larger than $\log \pi(\hat{\theta})$.

After the data is observed, we can compute the distribution of a future observation Y from the same model conditional on the data. This distribution is called the posterior predictive distribution. Marginalizing the joint posterior $Y, \theta|X$ or by the law of total

probability, it is obtained as

$$\begin{aligned} f(y \mid x) &= \int f(y, \theta \mid x) d\theta \\ &= \int f(y \mid x, \theta) \pi(\theta|x) d\theta \\ &= \int f(y \mid \theta) \pi(\theta|x) d\theta, \end{aligned}$$

where, in the last line, we assume that Y is independent of X conditional on θ.

Example 1.2 (Normal means, ctd.)**.** The part in the numerator which does not depend on θ was found to be

$$\exp\left(-\frac{1}{2}\left(\sum_{i=1}^{n}(x_i-\mu)^2 - \frac{n^2\tau^2}{1+n\tau^2}(\bar{x}-\mu)^2\right)\right).$$

Using $\mathbf{1} = (1, \ldots, 1)^T$, the quadratic form in the exponent can be written as

$$(x-\mu\mathbf{1})^T\left(I - \frac{\tau^2}{1+n\tau^2}\mathbf{1}\mathbf{1}^T\right)(x-\mu\mathbf{1}) = (x-\mu\mathbf{1})^T\left(I+\tau^2\mathbf{1}\mathbf{1}^T\right)^{-1}(x-\mu\mathbf{1}).$$

This shows that the prior predictive distribution of $(X_1, \ldots, X_n)$ is normal with mean $\mu\mathbf{1}$ and covariance matrix $\Sigma = I + \tau^2\mathbf{1}\mathbf{1}^T$. This can also be seen by writing $X_i = \theta + \varepsilon_i$ where $\theta \sim N(\mu, \tau^2)$, $\varepsilon_i \sim \mathcal{N}(0,1)$ and all variables are independent. Finally, if we write a new observation X_{n+1} as $\theta + \varepsilon_{n+1}$, it follows that the posterior predictive density is

$$X_{n+1} \mid (X_1 = x_1, \ldots, X_n = x_n) \sim \mathcal{N}\left(\frac{1}{1+n\tau^2}\mu + \frac{n\tau^2}{1+n\tau^2}\bar{x}, 1 + \frac{\tau^2}{1+n\tau^2}\right).$$

1.2.2 Point estimation and decision theory

As a Bayesian point estimate of θ, we can take any functional of location like expectation, median or mode of the posterior. If θ is a scalar, the posterior mean is

$$\widehat{\theta} = \mathbb{E}(\theta \mid x) = \int_{-\infty}^{\infty} \theta\pi(\theta \mid x)d\theta = \frac{\int_{-\infty}^{\infty}\theta\pi(\theta)f(x \mid \theta)d\theta}{\int_{-\infty}^{\infty}\pi(\theta)f(x \mid \theta)d\theta},$$

the posterior median is the solution of

$$\int_{-\infty}^{\widehat{\theta}} \pi(\theta)f(x \mid \theta)d\theta = \frac{1}{2}\int_{-\infty}^{\infty}\pi(\theta)f(x \mid \theta)d\theta$$

and the posterior mode is

$$\widehat{\theta} = \arg\max_{\theta} \pi(\theta \mid x) = \arg\max_{\theta} \left(\log\pi(\theta) + \log f(x \mid \theta)\right),$$

where arg max denotes the argument where a function takes its maximal value.

A unified approach which covers all these possibilities chooses a loss function

$$L : \Theta^2 \to [0, \infty)$$

and minimizes the posterior risk, i.e., the expected loss with respect to the posterior:

$$\widehat{\theta} = \arg\min_{T} \rho(T(x), \pi),$$

where

$$\rho(T(x), \pi) = \mathbb{E}(L(T(X), \theta) \mid x) = \int_\Theta L(T(x), \theta)\pi(\theta \mid x)d\theta.$$

Here, $L(T(x), \theta)$ quantifies our loss if the true value is θ and our estimate is $T(x)$. We call an estimator T that minimizes the posterior risk a Bayes estimator.[2] We obtain the posterior mean if $L(T, \theta) = (T - \theta)^2$, the posterior median if $L(T, \theta) = |T - \theta|$ and the posterior mode if $L(T, \theta) = 1_{[-\varepsilon,\varepsilon]^c}(T - \theta)$ and we let ε go to zero.

In the frequentist approach, a point estimate $T(x)$ is evaluated in terms of the frequentist risk which is the expected loss with respect to the distribution of x:

$$R(T, \theta) = \mathbb{E}_\theta(L(T(X), \theta)) = \int_{\mathbf{X}} L(T(x), \theta)f(x \mid \theta)dx$$

which depends on the unknown θ. We would like to minimize the risk for all θ simultaneously, but this is not possible because we can achieve a low risk for some value θ_0 if we are prepared to live with a high risk for values far away from θ_0 (consider the trivial estimator $T(x) \equiv \theta_0$, regardless of what the data are). To put it differently, if we have two estimators T_1 and T_2 which are both reasonable in an intuitive sense then we often have two values θ_1 and θ_2 such that

$$R(T_1, \theta_1) < R(T_2, \theta_1), \text{ but } R(T_1, \theta_2) > R(T_2, \theta_1).$$

In other words, whether T_1 or T_2 is better, depends on the unknown true value of the parameter θ.

In view of this dilemma, one solution is the minimax approach which looks for the estimator T which minimizes the maximal risk $\sup_\theta R(T, \theta)$.

A less pessimistic approach is to look for the estimator T which minimizes the the weighted risk

$$R(T, w) = \int_\Theta R(T, \theta)w(\theta)d\theta.$$

If $\int w(\theta)d\theta < \infty$, we can assume w.l.o.g. that w is a probability density. The next theorem shows that the estimator minimizing this weighted risk is then the Bayes estimator with prior $\pi = w$. For this reason, one also calls $R(T, w)$ the Bayes risk.

Theorem 1.1. *Assume that $\int w(\theta)d\theta = 1$ and choose w as the prior for θ. If*

$$T(x) = \arg\min_T \rho(T(x), w) = \arg\min_T \mathbb{E}(L(T, \theta) \mid x)$$

is well defined for almost all x with respect to the prior predictive distribution $f(x) = \int f(x \mid \theta)w(\theta)d\theta$, then T minimizes the weighted risk $R(T, w)$. Any other minimizer T' is almost surely equal to T.

Proof. By exchanging the order of integration, the weighted risk is

$$R(T, w) = \int_{\mathbf{X}} \left(\int_\Theta L(T(x), \theta)f(x|\theta)w(\theta)d\theta \right) dx$$

[2] In frequentist statistics, one usually calls an estimator that minimizes the weighted frequentist risk, the so-called Bayes risk, a Bayes estimator. However, the Bayesian estimator that minimizes the posterior risk and the frequentist estimator that minimizes the weighted risk are essentially equivalent. See below in the script for more details.

By Bayes' formula $f(x|\theta)w(\theta) = f(x)\pi(\theta \mid x)$ and therefore

$$R(T, w) = \int_{\mathbf{X}} \left(\int_{\Theta} L(T(x), \theta)\pi(\theta \mid x)d\theta \right) f(x)dx.$$

In order to minimize the right-hand side, we have to minimize the inner integral for almost all x. □

In particular, we can avoid the discussion whether it is legitimate to consider θ as random using the prior only to weight the risk for the different values of θ. The posterior is then just a technical device to compute the estimator which minimizes the weighted risk.

Another approach to address the problem that there is no estimator which minimizes the frequentist risk simultaneously for all θ is admissibility. An estimator T is called admissible if no other estimator T' exists which is uniformly better than T: If $R(T', \theta) \leq R(T, \theta)$ for all θ, then we must have $R(T', \theta) = R(T, \theta)$ for all θ. This allows to discard obviously bad estimators as inadmissible, but the class of admissible estimators is typically very large. One show that a Bayes estimator is admissible if the frequentist risk is continuous in θ for any estimator with finite risk and if the prior density is strictly positive everywhere. There is a large literature showing that under certain conditions any admissible estimator is a limit (in a sense to be made precise) of Bayes estimators. Such results give a justification of Bayes estimators from a frequentist point of view.

In the example of normal means the posterior mean $\widehat{\theta} = \frac{1}{1+n\tau^2}\mu + \frac{n\tau^2}{1+n\tau^2}\bar{x}$ is obviously biased:

$$\mathbb{E}_\theta(\widehat{\theta}) = \frac{1}{1+n\tau^2}\mu + \frac{n\tau^2}{1+n\tau^2}\theta \neq \theta \quad \forall \theta \neq \mu.$$

This is true in most cases because the choice of a prior usually means that not all values θ are considered equal. Since also modern frequentist statistics tends to deemphasize unbiasedness, this is however not a serious disadvantage of Bayesian estimators.

1.2.3 Testing, Bayes factor and credible sets

Because in Bayesian statistics the parameter is random, it becomes possible to speak about the "probability that the null hypothesis is true" or the "probability that θ belongs to some interval". In frequentist statistics, such statements have no meaning, and one has to be very careful if one wants to explain the meaning of a p-value or a confidence interval in words.

Let us first discuss Bayesian testing of a null hypothesis $\theta \in \Theta_0 \subset \Theta$ against the alternative $\theta \in \Theta_1 = \Theta_0^c$. The posterior probability of the null hypothesis is then

$$\pi(\Theta_0 \mid x) = \int_{\Theta_0} \pi(\theta \mid x)d\theta = \frac{\int_{\Theta_0} f(x \mid \theta)\pi(\theta)d\theta}{\int_{\Theta} f(x \mid \theta)\pi(\theta)d\theta}.$$

A Bayesian test will reject the null hypothesis iff $\pi(\Theta_0 \mid x)$ is below some threshold c. If we quantify the loss in case of an error of the first kind as a_1 and in case of an error of the second kind as a_2, then the posterior expected loss of a test $\varphi : \mathbf{X} \to \{0, 1\}$ is $a_2(1 - \pi(\Theta_0 \mid x))$ if $\varphi(x) = 0$ and $a_1\pi(\Theta_0 \mid x)$ if $\varphi(x) = 1$, where $\varphi(x) = 1$ means that the null hypothesis is rejected. Hence the posterior expected loss is minimized if $\varphi(x) = 0$ for $\pi(\Theta_0 \mid x) > a_2/(a_1 + a_2)$ and $\varphi(x) = 1$ for $\pi(\Theta_0 \mid x) \leq a_2/(a_1 + a_2)$, i.e. $c = a_2/(a_1 + a_2)$.

Instead of $\pi(\Theta_0 \mid x)$, in Bayesian statistics, one often considers the *Bayes factor* which is defined as the ratio of posterior and the prior odds in favor of the null hypothesis

$$B(x) = \frac{\pi(\Theta_0 \mid x)}{\pi(\Theta_1 \mid x)} \frac{\pi(\Theta_1)}{\pi(\Theta_0)}.$$

Note that $\pi(\Theta_0)$ and $\pi(\Theta_1)$ can be considered as (hyper)priors on the models themselves, and the overall prior can be written as

$$\pi(\theta) = \pi(\Theta_0)\pi_0(\theta) + \pi(\Theta_1)\pi_1(\theta),$$

where we use the following notation to denote the conditional priors under the null and the alternative

$$\pi_0(\theta) = \frac{\pi(\theta)1_{\Theta_0}(\theta)}{\pi(\Theta_0)}, \quad \pi_1(\theta) = \frac{\pi(\theta)1_{\Theta_1}(\theta)}{\pi(\Theta_1)}.$$

Using this notation, the Bayes factor can be written as

$$B(x) = \frac{\int_{\Theta_0} f(x \mid \theta)\pi_0(\theta)d\theta}{\int_{\Theta_1} f(x \mid \theta)\pi_1(\theta)d\theta} = \frac{f(x \mid \theta \in \Theta_0)}{f(x \mid \theta \in \Theta_1)}.$$

I.e., the Bayes factor is the ratio of the marginal densities, or marginal likelihoods, $f(x \mid \theta \in \Theta_0)$ and $f(x \mid \theta \in \Theta_1)$ of x under the null and the alternative hypothesis.

If $\Theta = \{\theta_0, \theta_1\}$, the Bayes factor is independent of the prior and equal to the likelihood ratio $f(x \mid \theta_0)/f(x \mid \theta_1)$ used in the Neyman-Pearson lemma. For composite hypotheses, the Bayes factor still depends on the prior.

A Bayes factor between 1 and $\frac{1}{3}$ is considered as weak evidence against the null, a value between $\frac{1}{3}$ and 0.1 as substantial, a value between 0.1 and 0.01 as strong and a value below 0.01 as decisive, according to Jeffreys (1961).

In some applications the null hypothesis consists of a subset with Lebesgue measure zero, typically a lower dimensional subset of Θ, e.g. in the case of $\mathcal{N}(\mu, \sigma^2)$-observations $\Theta_0 = \{(\mu, \sigma^2); \mu = \mu_0\}$. In this case, if we choose a prior which has a density with respect to the Lebesgue measure, the prior and the posterior give zero probability to the null hypothesis. Hence there would be no need to collect data as data cannot change the prior belief in Θ_0. In such situations one should therefore choose a prior which assigns to Θ_0 a probability strictly between 0 and 1. This can be achieved by a mixture

$$\pi(d\theta) = \rho_0\pi_0(d\theta) + (1 - \rho_0)\pi_1(\theta)d\theta,$$

where π_0 is a distribution which is concentrated on Θ_0 and ρ_0 is the prior probability of Θ_0. Note that because π_0 does not have a density with respect to the Lebesgue measure, we use here the general notation of measure theory. With such a prior, the posterior probability of Θ_0 is

$$\pi(\Theta_0 \mid x) = \frac{\rho_0 \int_{\Theta_0} f(x \mid \theta)\pi_0(d\theta)}{\rho_0 \int_{\Theta_0} f(x \mid \theta)\pi_0(d\theta) + (1 - \rho_0) \int_{\Theta} f(x \mid \theta)\pi_1(\theta)d\theta}.$$

Note that whether testing with point null hypotheses is reasonable in the first place or not is currently controversially debated.

In frequentist statistics, the p-value is sometimes taken as a measure of evidence against the null hypothesis. It is defined as the smallest significance level for which the null hypothesis is still rejected under the assumption that the null hypothesis holds true. Although

conceptually this is not the same as the posterior probability of the null hypothesis, it would be nice if these two measures were close at least in the case where the null and the alternative are a priori equally likely. Let us consider the case where $\Theta_0 = \{\theta_0\}$. Then for $\rho_0 = \frac{1}{2}$, we have

$$\pi(\Theta_0 \mid x) = \frac{f(x \mid \theta_0)}{f(x \mid \theta_0) + \int_\Theta f(x \mid \theta)\pi_1(\theta)d\theta}$$

which depends on the chosen prior for the alternative, but there is the trivial lower bound

$$\inf_{\pi_1} \pi(\Theta_0 \mid x) = \frac{f(x \mid \theta_0)}{f(x \mid \theta_0) + \sup_\theta f(x \mid \theta)}.$$

p-value	0.10	0.05	0.01	0.001
$\inf_{\pi_1} \pi(\Theta_0 \mid x)$	0.205	0.128	0.035	0.004
$\inf_{\pi_1} B(x)$	0.256	0.146	0.036	0.004
$\inf_{\pi_1 \in \mathcal{S}} \pi(\Theta_0 \mid x)$	0.392	0.290	0.109	0.018
$\inf_{\pi_1 \in \mathcal{S}} B(x)$	0.644	0.409	0.123	0.018

Table 1.1: Testing the null hypothesis $\mu = \mu_0$ for i.i.d. normal observations with mean μ and known variance: Comparison of p-value and lower bounds on the posterior probability for the null under arbitrary (rows 2) and under symmetric unimodal (row 4) priors for the alternative. The prior probability of the null hypothesis is equal to $\frac{1}{2}$. Source: Tables 4 and 6 in Berger and Selke, JASA 82 (1987)

In many situations this lower bound is substantially larger than the corresponding p-value. See, e.g., Table 1.1 where p-values are compared with corresponding posterior probabilities (row 2) when testing the null hypothesis $\mu = \mu_0$ for i.i.d. normal observations with mean μ and known variance. Posterior probabilities (and Bayes factors) are calculated using the same values of the t-statistics that result in the corresponding p-values. This example shows that the p-value overestimates the evidence against the null even when the prior is heavily biased towards the alternative. If one assumes θ to be scalar and restricts π_1 to the class $\mathcal{S}$ of symmetric unimodal densities, then one can show that

$$\inf_{\pi_1 \in \mathcal{S}} \pi(\Theta_0 \mid x) = \frac{f(x \mid \theta_0)}{f(x \mid \theta_0) + \sup_c \frac{1}{2c} \int_{\theta_0 - c}^{\theta_0 + c} f(x \mid \theta)d\theta}.$$

The discrepancy between the posterior probability of the null and the p-value is now even more drastic; see the fourth row of Table 1.1.

A Bayesian confidence set with level $1 - \alpha$ is called a $(1 - \alpha)$-*credible set*. It is a subset $C_x \subset \Theta$ (depending on x) such that

$$P(\theta \in C_x \mid X = x) = \pi(C_x \mid x) \geq 1 - \alpha.$$

For one dimensional parameters, credible sets are also called credible intervals. For a Bayesian credible set, x is fixed and θ is random, whereas in the frequentist approach θ is fixed and x is random and we require

$$P_\theta(C_x \ni \theta) \geq 1 - \alpha \quad \forall \theta \in \Theta.$$

Note that the different orders $\theta \in C_x$ and $C_x \ni \theta$ are chosen to emphasize what is random and what is fixed.

Among the many $(1-\alpha)$-credible sets, the one minimizing the volume (Lebesgue measure) is particularly attractive. It is obtained by taking C_x as a level set of the posterior, $C_x = L_{k_\alpha}$, where

$$L_k = \{\theta; \pi(\theta \mid x) \geq k\}, \quad k_\alpha = \sup\{k; \pi(L_k \mid x) \geq 1-\alpha\}.$$

It is thus called a highest posterior density (HPD) credible set. That it minimizes the volume can be seen as follows. For simplicity, we assume that $\pi(L_{k_\alpha} \mid x) = 1-\alpha$. If C is another $(1-\alpha)$-credible set, then by the definition of the level set

$$\begin{aligned} 0 &\geq \pi(L_{k_\alpha} \mid x) - \pi(C \mid x) = \int_{L_{k_\alpha} \cap C^c} \pi(\theta \mid x) d\theta - \int_{L_{k_\alpha}^c \cap C} \pi(\theta \mid x) d\theta \\ &> k_\alpha \left(|L_{k_\alpha} \cap C^c| - |L_{k_\alpha}^c \cap C| \right) \end{aligned}$$

where $|C|$ denotes the volume (Lebesgue measure) of a set C. In high dimensions, the computation of L_{k_α} can be difficult.

For continuous one dimensional parameters, the central credible interval (c_l, c_u) is defined such that

$$P(\theta < c_l \mid X = x) = \alpha/2 \quad \text{and} \quad P(\theta > c_u \mid X = x) = \alpha/2.$$

Figure 1.1 shows an example of a one-dimensional highest posterior density (HPD) credible interval and a central credible interval.

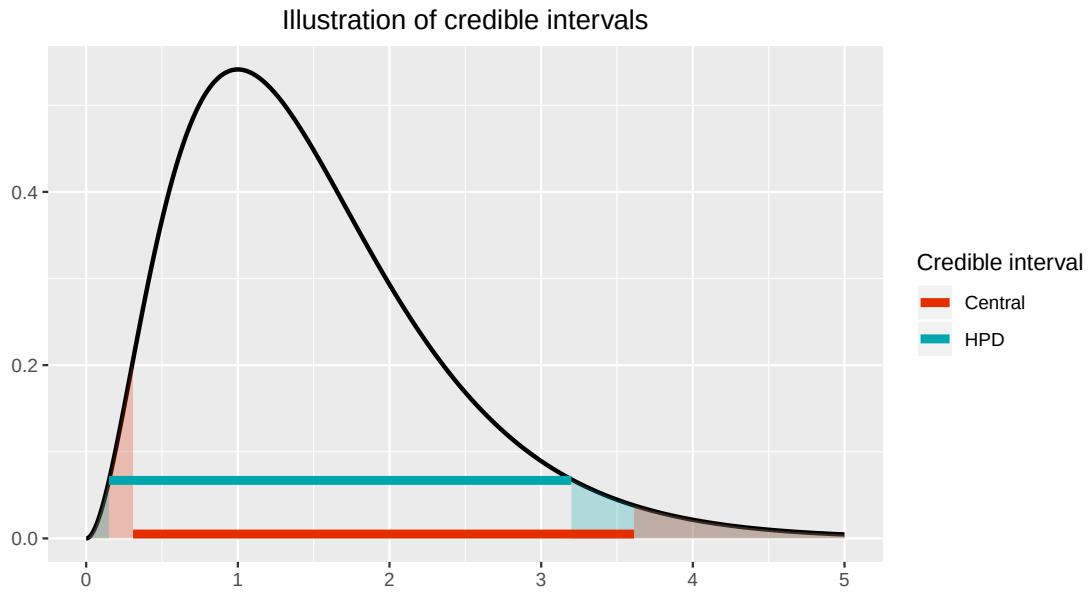


Figure 1.1: Illustration of a one-dimensional highest posterior density (HPD) credible interval and a central credible interval.

1.2.4 Bayesian asymptotics

Two important results in frequentist statistics are the following ones. Assume an i.i.d. model $X_i \sim f(x \mid \theta)dx$ where $\theta \in \Theta$ and Θ is an open set in $\mathbb{R}^p$ and denote the likelihood function by $L_n(\theta) = \prod_{i=1}^n f(x_i \mid \theta)$, the maximum likelihood estimator (MLE) by $\widehat{\theta}_n = \arg\max_\theta L_n(\theta)$ and the Fisher information by

$$I(\theta) = -\mathbb{E}_\theta \left(\frac{\partial^2}{\partial\theta\partial\theta^T} \log f(X_i \mid \theta) \right).$$

Then, under regularity conditions, if θ_0 is the true parameter,

$$\widehat{\theta}_n \overset{\text{approx}}{\sim} \mathcal{N}\left(\theta_0, \frac{1}{n} I(\theta_0)^{-1}\right)$$

and

$$2\left(\log L_n(\widehat{\theta}_n) - \log L_n(\theta_0)\right) \xrightarrow{d} \chi^2_p.$$

Bayesian asymptotics, i.e. the Bernstein-von Mises theorem, says that for any prior which is strictly positive and continuous in a neighborhood of θ_0

$$\theta \mid (x_1, \ldots, x_n) \overset{\text{approx}}{\sim} \mathcal{N}\left(\widehat{\theta}_n, \frac{1}{n} I(\widehat{\theta}_n)^{-1}\right).$$

More precisely, under regularity conditions, the following holds true:

$$\left\| \pi\left(\theta \mid (x_1, \ldots, x_n)\right) - \mathcal{N}\left(\widehat{\theta}_n, \frac{1}{n} I(\widehat{\theta}_n)^{-1}\right) \right\|_{TV} \xrightarrow{P_{\theta_0}} 0$$

where

$$TV(P, Q) = \sup_{B \in \mathcal{B}(\mathbb{R}^p)} |P(B) - Q(B)|$$

and $\mathcal{B}(\mathbb{R}^p)$ is the Borel σ-algebra on $\mathbb{R}^p$. Therefore, the influence of the prior disappears asymptotically and the posterior is concentrated in a $\sqrt{1/n}$ neighborhood of the MLE. There is a nice symmetry in the asymptotic statements, but note again the difference in what is considered fixed and what is random in the two approaches. See, for instance, van der Vaart (2000), Theorem 10.1, or Schervish (1995), Chap. 7.4, for a proof.

Figure 1.2 illustrates the Bernstein-von Mises theorem. The figure shows the posterior and the asymptotic approximation for a beta-binomial model with $\alpha = \beta = 0.5$; see Section 2.1.

1.3 Interpretations of probability

In Bayesian statistics, one often interprets probabilities as quantification of uncertainty. In the following, we first elaborate on this. Jakob Bernoulli wrote "Probability is the degree of certainty which is to the certainty as a part is to a whole". There are however two somewhat different kinds of (un)certainty, called "aleatoric uncertainty" and "epistemic uncertainty". Aleatoric uncertainty (from Latin "alea" = dice) is the uncertainty about outcomes of repeatable events like throwing a dice. In such cases, probabilities can be understood as mathematical idealizations of long-run relative frequencies, which we call the frequentist interpretation of probability. Epistemic uncertainty (from Greek "episteme" = knowledge) is the uncertainty about unique events resulting from insufficient knowledge, e.g. about whether a particular student will pass an exam or whether the increased release of CO_2 is the reason for the increasing global temperatures in the past 50 years. It does not make sense to imagine that a student takes the same exam many times, or that there are other earth systems where humans release CO_2 into the atmosphere. In such cases, the probability statements are based on an evaluation of the available facts and information by an individual and thus become subjective degrees of belief. Hence this approach is called the subjective or Bayesian approach (after Thomas Bayes, c.1701-1761). Subjectivity does not go well together with our ideal of scientific objectivity and thus this interpretation of probability is often criticized or rejected. However, subjectivity is not the same thing as

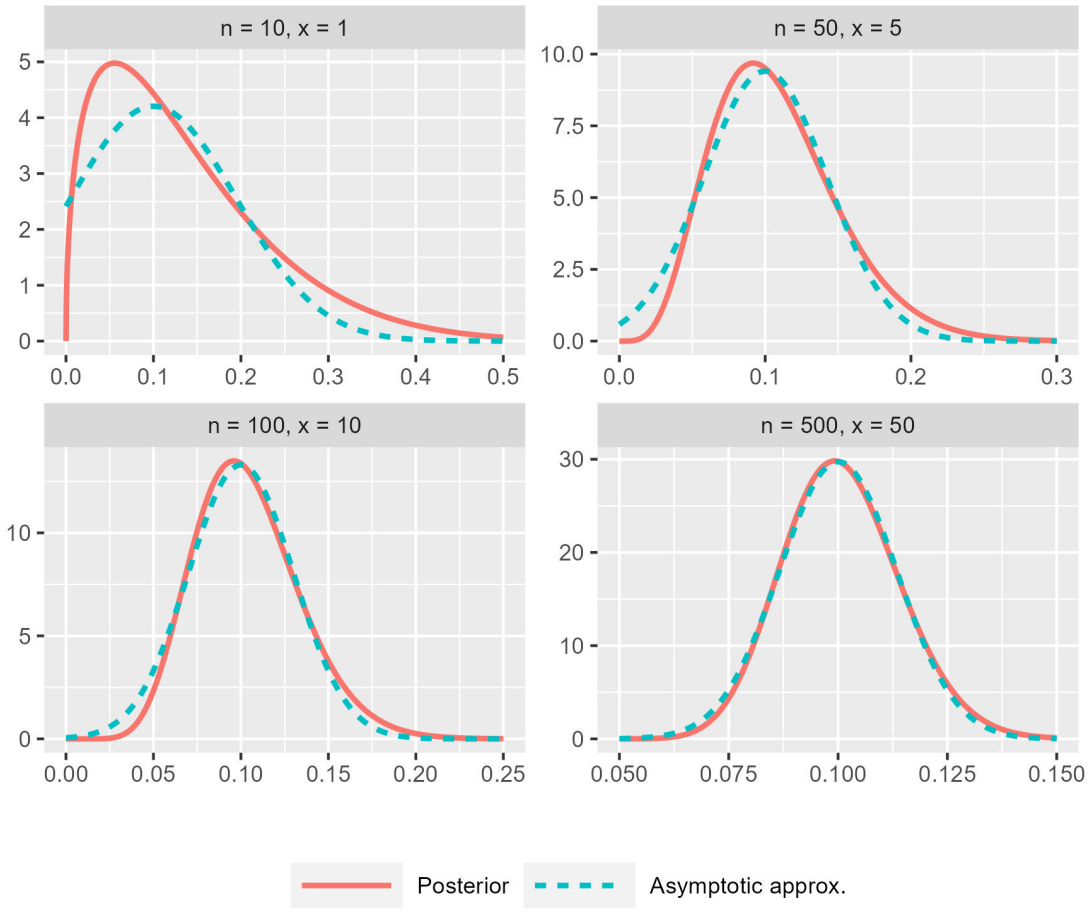


Figure 1.2: Illustration of Bernstein-von Mises theorem: posterior and asymptotic approximation for a beta-binomial model with $\alpha = \beta = 0.5$ and varying sample sizes n.

arbitrariness – an individual should be able to reveal and explain the information on which his or her assignment of probabilities is based. Furthermore, most scientific models which rely on mathematical idealizations of the world are subjective to some degree. A famous quote attributed to George Box about statistical models is that "all models are wrong, but some are useful".

One can argue that aleatoric uncertainty is also a form of epistemic uncertainty about a physical system: If we knew exactly the position and the momentum of the dice when it leaves the hand of the player, then we should be able to predict the result. This is however not particularly helpful, and the distinction between repeatable and unique events remains.

One may ask whether we should measure the two types of uncertainty by the same concept. There is however an argument due to de Finetti showing that in order to act rationally we should measure epistemic uncertainties by probabilities.

This argument goes as follows. Assume that we have a finite set Ω of outcomes, and that there is a bookmaker who has to offer bets on all subsets A of Ω other than Ω itself and the empty set. He has to choose the payment odds λ_A which means that a gambler who bets $b_A \in \mathbb{R}$ francs on A will win b_A/λ_A franks (in addition to the stake b_A that is returned) if A occurs and $-b_A$ francs if A does not occur. The stake b_A of the gambler can be negative: Because we do not make any assumptions on the payment odds, betting

a negative amount on A need not be equivalent to betting a positive amount on A^c. The net payoff of the gambler betting b_A on A is therefore

$$\phi_A(\omega) = 1_A(\omega)\frac{b_A}{\lambda_A} - (1 - 1_A(\omega))b_A = b_A\frac{1+\lambda_A}{\lambda_A}\left(1_A - \frac{\lambda_A}{1+\lambda_A}\right).$$

Clearly λ_A should reflect the bookmakers epistemic uncertainty about the occurrence of A: The higher his belief is in the occurrence of A, the larger λ_A should be. We say that the bookmaker is a 'Bayesian with prior π', where π is a probability measure on Ω, if he chooses his odds according to

$$\lambda_A = \pi(A)/(1 - \pi(A))$$

for all A.

The following theorem due to de Finetti shows that if the bookmaker does not assign his payoff odds according to some probability π on Ω, then a gambler can combine bets on different events A in such a way that the total payoff is strictly positive no matter what the outcome ω is. So we can deduce the subjective probabilities of a bookmaker from the payment odds he offers.

Theorem 1.2.

(i) If the bookmaker is a Bayesian with prior π, then for any choice of bets (b_A), the payoff function

$$V(\omega) = \sum_{A\subset\Omega,\ A\neq\emptyset,\Omega} \phi_A(\omega)$$

has expectation zero with respect to π. In particular, it takes both positive and negative values.

(ii) If the bookmaker is not a Bayesian, then for any function $V : \Omega \to \mathbb{R}$ there is a choice of bets (b_A) such that

$$V(\omega) = \sum_{A\subset\Omega,\ A\neq\emptyset,\Omega} \phi_A(\omega).$$

In particular, a gambler can place bets such that the payoff is equal to one for any outcome ω.

Proof. The first statement follows by straightforward computation. For the second part, we use that the set of possible payoff functions is a linear subspace of $\mathbb{R}^{|\Omega|}$, namely the space spanned by all functions ϕ_A. If this subspace is not the whole space, then there is a nonzero vector π such that for all A

$$\sum_{\omega} \pi(\omega)\phi_A(\omega) = 0.$$

By the definition of ϕ_A, this means that for all A

$$\frac{1+\lambda_A}{\lambda_A}\sum_{\omega\in A}\pi(\omega) = \sum_{\omega\in\Omega}\pi(\omega).$$

By choosing as A all one-point sets, we see that $\sum_{\omega\in\Omega}\pi(\omega) \neq 0$ because otherwise π would be the zero vector. Without loss of generality, we assume that $\sum_{\omega\in\Omega}\pi(\omega) = 1$. Choosing

again as A all one-point sets, we also obtain $\pi(\omega) > 0$ for all ω so that π is indeed a probability. Moreover,

$$\pi(A) = \sum_{\omega \in A} \pi(\omega) = \frac{\lambda_A}{1 + \lambda_A}.$$

□

This theorem does not tell which probability π a Bayesian bookmaker should use to fix the payments odds. If we are in a situation with repeated outcomes according to some probability P, then in the long run a gambler can make a profit with high probability if $\pi \neq P$. But in such a situation a Bayesian bookmaker would adjust the payment odds and thus also the underlying π based on past observations.

In the case where Ω is infinite and one has to show σ-additivity of π, there are some technical complications that we do not discuss here. See the unpublished note by D. A. Freedman, "Notes on the Dutch Book Argument" (2003), available at `http://www.stat.berkeley.edu/~census/` on which the theorem of this section is based.

1.4 Likelihood principle

We have seen in this chapter that a basic difference between the frequentist and the Bayesian approach is that the former considers other values of the data that did not occur, but could have been obtained, whereas the latter considers only the data that were actually observed. To justify the frequentist approach, we mention the following quote from Mosteller and Tukey (1968): "One hallmark of the statistically conscious investigator is his firm belief that however the survey, experiment or observational program actually turned out, it could have turned out somewhat differently. Holding such a belief and taking appropriate actions make effective use of data possible. We need not always ask explicitly "How much differently?", but we should be aware of such questions."

On the other hand, there are clearly situations where considering data that were not, but might have been obtained leads to strange conclusions: Consider the estimation of an unknown concentration of a substance in a probe. The probe can be analyzed by two labs, one which measures with high precision and one with low precision. The high precision lab is however not always available, due to high demand from other customers. Let us assume that the standard deviations of both labs are known and equal to 1 and 10 respectively, and that the chance the precise lab is available is 0.5. If the analysis was made by the imprecise lab, then we only know that the true value is within ± 19.6 from the result. Arguing that because there was a 50% chance to have the analysis done in the precise lab and that therefore the standard deviation is $\sqrt{0.5 \cdot 1^2 + 0.5 \cdot 10^2} = 7.1$ is obviously not reasonable.

This can be formulated as the
Conditionality principle: If an experiment for inference about a parameter θ is chosen independently from a collection of different possible experiments, then any experiment not chosen is irrelevant to the inference.

This principle seems quite indisputable. There is another principle which is also not controversial, namely that observations which differ only in a way which is irrelevant for the model under consideration should lead to the same conclusion. For instance, in a model where the observations X_i are i.i.d. $\sim f(x|\theta)dx$, the time order of the observations is irrelevant. The time order can give information whether the i.i.d. assumption is reasonable

or not, but once we decide to use the model, the time order of the observation does not give any information about θ. In mathematical statistics this idea is formalized by the concept of a *sufficient statistic*: A function $T(x)$ of the observation x is sufficient for a model $(f(x|\theta); \theta \in \Theta)$ if the conditional distribution of x given $T(x) = t$ does not involve θ. Then we can formulate the following principle.

Sufficiency principle: If there are two observations x and y such that $T(x) = T(y)$ for a sufficient statistic T, then any conclusion about θ should be the same for x and y.

The surprising result due to Birnbaum (1962) is that the two largely non-controversial principles above imply a third principle which is violated by many frequentist tests and confidence intervals:

Likelihood principle: If there are two different experiments for inference about the same parameter θ and if the outcomes x and y from the two experiments are such that the likelihood functions differ only by a multiplicative constant, then the inference should be the same.

To understand this principle, let us consider the problem where there is some event with unknown probability p and we want to test the null hypothesis $p \leq 0.5$ against the alternative $p > 0.5$. This can be done either by repeating the trial n times and observing the number X of trials where the event occurred, or by repeating the experiment a random number N times until the event has occurred a fixed number of times x. In the first experiment X is random, n is fixed,

$$P_p(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

and and the null hypothesis is rejected if $x > c_1$ which is equivalent to $x/n > c_1'$, where $c_1' = c_1 \cdot n$ (n is fixed). In the second experiment N is random, x is fixed and

$$P_p(N = n) = \binom{n-1}{x-1} p^x (1-p)^{n-x}$$

because the last occurrence of the event must be in the n-th trial, but the other $x-1$ occurrences can be any time before the n-th trial. In the second experiment, we will reject the null if $n < c_2$ which is equivalent to $x/n > x/c_2 =: c_2'$ (remember that x is fixed). Hence the form of the rejection region is the same in both experiments, but the boundary values c_1' and c_2' differ in general because they are computed differently:

$$\sum_{k=c_1}^{n} \binom{n}{k} 2^{-n} > \alpha \geq \sum_{k=c_1+1}^{n} \binom{n}{k} 2^{-n},$$

but

$$\sum_{m=x}^{c_2-1} \binom{m-1}{x-1} 2^{-m} \leq \alpha < \sum_{m=x}^{c_2} \binom{m-1}{x-1} 2^{-m}.$$

Hence for a frequentist test it matters whether we observe x successes in n trials in experiment 1 or experiment 2 although the likelihood functions $p \mapsto P_p(X = x)$ and $p \mapsto P_p(N = n)$ are in both cases proportional to $p^x(1-p)^{n-x}$. In particular, the Bayes test gives the same answer regardless which experiment was performed.

Note that the maximum likelihood estimator is x/n, regardless whether experiment 1 or 2 was made: Point estimation by maximum likelihood does obey the likelihood principle.

For a proof of Birnbaum's result, see for instance Section 1.3.3 in Robert's book.

Chapter 2

Prior distributions

The choice of a prior is a point which has lead to an intensive debate and which is often considered to be the weak point of the Bayesian approach. We discuss three approaches. The first one chooses prior distributions such that the posterior can be easily computed. The second one tries to determine a prior which contains as little information as possible. The third one tries to choose a prior based on the opinion of one or several experts.

2.1 Conjugate priors

In the example 1.1 a normal prior lead to a normal posterior and therefore the application of Bayes' formula becomes particularly simple: We need to know only how to compute the mean and variance of the posterior. The following definition generalizes this feature.

Definition 2.1. *A parametric family $\mathcal{P}_\Xi = \{\pi_\xi(\theta); \xi \in \Xi\}$, $\Xi \subset \mathbb{R}^q$, of prior densities is called* conjugate *for the model $\{f(x \mid \theta); \theta \in \Theta\}$ if, for any $\pi \in \mathcal{P}_\Xi$ and any x, $\pi(\theta \mid x)$ is again in $\mathcal{P}_\Xi$.*

Written out, this means that to any $\xi \in \Xi$ and any x there must be a $\xi' = \xi'(\xi, x)$ such that

$$\pi_\xi(\theta) f(x \mid \theta) \propto \pi_{\xi'}(\theta).$$

Computing the posterior amounts then to computing $\xi'(\xi, x)$.

It is obvious that $\mathcal{P}_\Xi$ is conjugate if the following two conditions are satisfied

1. To any x there is a $\xi(x) \in \Xi$ such that $f(x \mid \theta) \propto \pi_{\xi(x)}(\theta)$.
2. To any pair $\xi_1, \xi_2 \in \Xi$ there is a $\xi_3 \in \Xi$ such that $\pi_{\xi_1}(\theta)\pi_{\xi_2}(\theta) \propto \pi_{\xi_3}(\theta)$.

Example 2.1. Consider the binomial distribution

$$f(x \mid \theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x}.$$

It is clear that the above conditions are satisfied if we choose $\pi_\xi(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$ where $\xi = (\alpha, \beta) \in \mathbb{N}^2$ or $\xi \in (0, \infty)^2$.

A class of conjugate priors $\mathcal{P}_\Xi$ remains conjugate under repeated sampling, i.e., it is also conjugate for the model where $X_1, \ldots, X_n$ are i.i.d., $X_i \sim f(x \mid \theta)dx$ and n is arbitrary because, for instance,

$$\pi(\theta \mid x_1, x_2) \propto \pi(\theta \mid x_1) f(x_2 \mid \theta).$$

Therefore, if $\mathcal{P}_\Xi$ is conjugate for $f(x \mid \theta)$, for arbitrary, but fixed ξ_0 we can write

$$\prod_{i=1}^{n} f(x_i \mid \theta) = \frac{\pi_{\xi_n(x_1,\ldots,x_n)}(\theta)}{\pi_{\xi_0}(\theta)} f_n(x_1, \ldots, x_n)$$

where f_n is the prior predictive density of $X_1, \ldots, X_n$ and where ξ_n maps n-tupels of observed values to Ξ. In the language of mathematical statistics, ξ_n is then a sufficient statistic whose dimension is the same for any n. The existence of a sufficient statistics of finite dimension has been studied in mathematical statistics, and it exists only for a restricted class of models, thus limiting the use of conjugate priors to such models. If the set $\{x; f(x \mid \theta) > 0\}$ does not depend on θ, f must belong to a so-called exponential family where the densities have the following form

$$f(x \mid \theta) = \exp(c_1(\theta)T_1(x) + \ldots + c_q(\theta)T_q(x) + d(\theta))h(x).$$

The conjugate family consists then of densities

$$\pi_\xi(\theta) \propto \exp(c_1(\theta)\xi_1 + \ldots + c_q(\theta)\xi_q + d(\theta)\xi_{q+1}).$$

The table below reports examples of exponential families together with their conjugate prior distributions.

Model	Prior	Posterior
Binomial(n, θ)	Beta(α, β)	Beta$(\alpha + x, \beta + n - x)$
Multinomial $(n, \theta_1, \ldots, \theta_k)$	Dirichlet$(\alpha_1, \ldots, \alpha_k)$	Dirichlet$(\alpha_1 + x_1, \ldots, \alpha_k + x_k)$
i.i.d. Poisson(θ)	Gamma(γ, λ)	Gamma$(\gamma + \sum_i x_i, \lambda + n)$
i.i.d. Normal$(\mu, \frac{1}{\tau})$ $\theta = (\mu, \tau)$	Normal$(\mu_0, \frac{1}{n_0\tau})$ $\times$ Gamma(γ, λ)	Normal$(\frac{n}{n+n_0}\bar{x} + \frac{n_0}{n+n_0}\mu_0, \frac{1}{(n+n_0)\tau})$ $\times$ Gamma$(\gamma + \frac{n}{2}, \lambda + \frac{1}{2}\sum_i (x_i - \bar{x})^2 + \frac{nn_0}{2(n+n_0)}(\bar{x} - \mu_0)^2)$
Uniform$(0, \theta)$	Pareto(α, σ)	Pareto$(\alpha + n, \max(\sigma, x_1, \ldots, x_n))$

Table 2.1: Most frequently used models with their conjugate priors

The distributions which appear in this table are defined as follows.

- Beta(α, β) where $\alpha > 0$ and $\beta > 0$ is the distribution on $(0, 1)$ with the density

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1 - x)^{\beta-1}.$$

  Here $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}dx$ is the Gamma function.

- Gamma(γ, λ) where $\gamma > 0$ and $\lambda > 0$ is the distribution on $(0, \infty)$ with the following density

$$f(x) = \frac{\lambda^\gamma}{\Gamma(\gamma)} x^{\gamma-1}e^{-\lambda x}.$$

  γ is often called the shape parameter and λ the rate parameter, and $1/\lambda$ is called the scale parameter.

- Inverse-gamma IG(γ, λ) where $\gamma > 0$ and $\lambda > 0$ is the distribution on $(0, \infty)$ with the following density
$$f(x) = \frac{\lambda^\gamma}{\Gamma(\gamma)} x^{-\gamma-1} e^{-\frac{\lambda}{x}}.$$
The parameter γ is called the shape parameter and λ the scale parameter.

- Multinomial$(n, p_1, \ldots, p_k)$ where $p_i \geq 0$, $\sum_i p_i = 1$ is the discrete distribution on the set $\{(x_1, \ldots, x_k) : x_i \geq 0, \sum_{i=1}^k x_i = n\}$ given by
$$p(x_1, \ldots, x_k) = \frac{n!}{(x_1)! \ldots (x_k)!} p_1^{x_1} \ldots p_k^{x_k}.$$

- Dirichlet$(\alpha_1, \ldots, \alpha_k)$ is the distribution on the simplex $\{(p_1, \ldots, p_{k-1}); p_i \geq 0, \sum_i p_i \leq 1\}$ with density
$$f(p_1, \ldots, p_{k-1}) = \frac{\Gamma(\alpha_1 + \ldots, \alpha_k)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_k)} p_1^{\alpha_1 - 1} \cdots p_{k-1}^{\alpha_{k-1}-1} (1 - p_1 - \ldots p_{k-1})^{\alpha_k - 1}.$$
For $k = 2$ this is simply the Beta-distribution.

- Pareto(α, σ) $(\alpha > 0, \sigma > 0)$ is the distribution on $[\sigma, \infty)$ with density
$$f(x) = \alpha \sigma^\alpha x^{-(\alpha+1)}.$$

In the case of the normal distribution, the bivariate prior for the mean μ and the inverse of the variance, the so-called precision τ is specified as conditional distribution of μ given τ (the first line in the table) times the marginal of τ (the second line in the table). Written out, the joint prior density is
$$\pi(\mu, \tau) = \frac{\sqrt{n_0 \tau}}{\sqrt{2\pi}} \exp\left(-\frac{n_0 \tau}{2}(\mu - \mu_0)^2\right) \frac{\lambda^\gamma}{\Gamma(\gamma)} \tau^{\gamma-1} \exp(-\lambda\tau).$$
The distribution with this density is called the normal-gamma distribution. Note that instead of assuming a gamma distribution for the precision τ, one can also assume an inverse-gamma distribution for the variance σ^2. Since $\tau = \sigma^{-2}$, these two priors are equivalent. The formula for the posterior follows therefore by multiplying the prior with the likelihood and using the same kind of algebraic manipulations as in Example 1.1. The marginal prior $\pi(\mu)$ for μ is a shifted and scaled t-distribution with 2γ degrees of freedom. This can be shown by integrating $\pi(\mu, \tau)$ over τ. Similarly, the marginal posterior of μ is a shifted and scaled t-distribution with $2\gamma + n$ degrees of freedom.

There are also analogous conjugate distributions for the d-dimensional normal distribution $\mathcal{N}_d(\mu, \Psi^{-1})$ with unknown mean vector μ and unknown precision matrix Ψ. The marginal of Ψ is then the so-called Wishart distribution, and the conditional distribution of μ given Ψ is multivariate normal. We refer to the literature for the definitions and the formulae.

Conjugate priors have again parameters, usually called *hyperparameters*, which have to be chosen. Note that, in the general formula, we denote hyperparameters by ξ, but in Table 2.1 different symbols are used. Hence using a conjugate prior does not fully answer the question "which prior ?" As typically there are more hyperparameters than parameters, choosing a hyperparameter seems even more difficult than choosing a parameter value. Note, however, that usually one of the hyperparameters can be regarded as a "hypothetical sample size" of the prior. In the general case, this is the parameter ξ_{q+1}. In the beta-binomial case it is $\alpha + \beta - 2$, in the gamma-Poisson case it is λ, and in the normal case

it is n_0 for μ and γ for τ. So this parameter can be determined by asking how much weight should be given the prior. The other parameters usually are related to a location parameter of the prior which helps to choose its value. For instance in the case of a Beta distribution, the mean is $\alpha/(\alpha+\beta)$.

2.2 Non-informative priors

The search for a non-informative prior is motivated by the wish to reduce the subjective element in a Bayesian analysis. A first attempt defines a uniform prior on Θ to be non-informative: This has however two drawbacks. First, the uniform distribution on Θ is a probability distribution only if Θ has finite volume. Second, the uniform distribution is not invariant under different ways to parametrize the same family of distribution. Assume we use $\tau = g(\theta)$ as our new parameter where g is invertible and smooth, i.e. $\theta = g^{-1}(\tau)$. If θ has density π, then by the change-of-variables formula, $\tau = g(\theta)$ has the density

$$\begin{aligned}\lambda(\tau) &= \pi(g^{-1}(\tau))|\det Dg(g^{-1}(\tau))|^{-1} \\ &= \pi(g^{-1}(\tau))|\det Dg^{-1}(\tau)|,\end{aligned}$$

where Dg is the Jacobi matrix of g whose (ij)-th entry is $\partial g_i/\partial\theta_j$, Dg^{-1} is the Jacobi matrix of g^{-1}, and we have use the fact that $Dg^{-1}(\tau)$ is the inverse matrix of $Dg(g^{-1}(\tau))$. This shows that if π is constant and g is not linear, then λ is not constant.

2.2.1 Improper priors

We call a (σ-finite) measure π on Θ which is not a probability measure, i.e.,

$$\int_\Theta \pi(\theta)d\theta = \infty,$$

an improper prior.

If π has infinite total mass, $\pi(\theta)f(x \mid \theta)$ can have both finite or infinite total mass, depending on the likelihood. If the total mass is finite, then we have by formal analogy the posterior density

$$\pi(\theta \mid x) = \frac{\pi(\theta)f(x \mid \theta)}{\int \pi(\theta')f(x \mid \theta')d\theta'}$$

and we can construct Bayesian point estimates, tests and credible intervals as before. Typically, this can be justified by approximating the improper prior by a sequence of proper priors π_k and showing that the associated sequence of posteriors $\pi_k(\theta \mid x)$ converges to the above expression. However, even if this convergence holds, paradoxes can occur. Moreover, in complicated models it is not always easy to check whether $\pi(\theta)f(x \mid \theta)$ has finite total mass.

2.2.2 Jeffreys prior

Jeffreys proposed to take

$$\pi(\theta) \propto \det(I(\theta))^{1/2}$$

where $I(\theta)$ is the Fisher information matrix that we already encountered in Section 1.2.4. It is defined as

$$I(\theta) = \mathbb{E}_\theta\left(\frac{\partial}{\partial\theta}\log f(X \mid \theta)\left(\frac{\partial}{\partial\theta}\log f(X \mid \theta)\right)^T\right),$$

and one can show that, under some regularity conditions, it is equal to

$$I(\theta) = -\mathbb{E}_\theta \left(\frac{\partial^2}{\partial\theta\partial\theta^T} \log f(X \mid \theta) \right).$$

The following two examples illustrate how Jeffreys prior is calculated for two specific models.

Example 2.2. Normal distribution. If $X \sim \mathcal{N}(\theta, 1)$, then $I(\theta) = 1$, so Jeffreys prior is the improper uniform distribution on $\mathbb{R}$. If $X \sim \mathcal{N}(0, \theta^2)$, $I(\theta) = 2/\theta^2$, so Jeffreys prior is $\pi(\theta) \propto \theta^{-1}$, which is again improper. It is the uniform distribution for the parameter $\tau = \log\theta$. If both mean and standard deviation are unknown, $X \sim \mathcal{N}(\mu, \sigma)$ and $\theta = (\mu, \sigma)$, then the Fisher information is diagonal with elements σ^{-2} and $2\sigma^{-2}$ and therefore Jeffreys prior is proportional to σ^{-2}. In particular, it is not the product of the univariate Jeffreys priors for μ and σ, contrary to what one would expect.

Example 2.3. Binomial distribution. If X is binomial(n, θ), then

$$I(\theta) = \mathbb{E}_\theta \left(\frac{X}{\theta^2} + \frac{n - X}{(1-\theta)^2} \right) = \frac{n}{\theta(1-\theta)}$$

so Jeffreys prior is the Beta$(\frac{1}{2}, \frac{1}{2})$ and not the uniform distribution. In particular it is proper.

Intuitively, the rationale for Jeffreys prior is as follows. Since $I(\theta)^{-1}$ is the asymptotic variance of the MLE, $I(\theta)$ can be seen as an indicator of the amount of information of the data about θ. If the prior also gives larger weight to values of θ for which the information in the data $I(\theta)$ is larger, the influence of the prior on the posterior is small and the prior is therefore non-informative. Furthermore, as we show in the following and as illustrated in Figure 2.1, Jeffreys prior is equivariant under different parametrizations.[1]

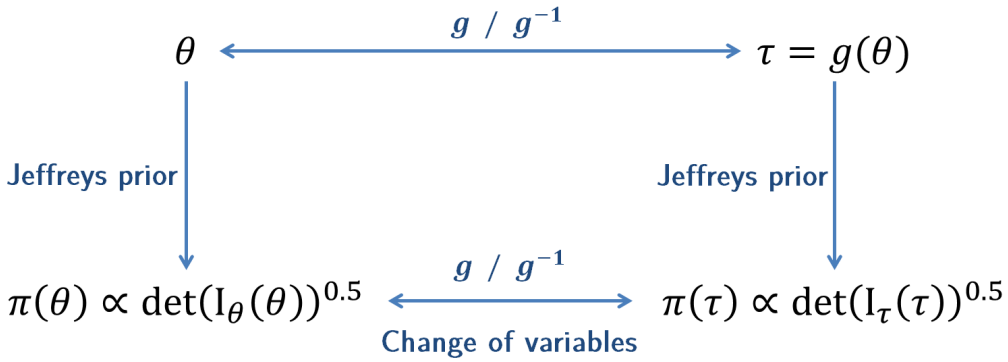


Figure 2.1: Equivariance of Jeffreys prior.

We first show the equivariance of Jeffreys prior in an example. We consider $X \sim \mathcal{N}(0, \sigma^2)$ and the four parametrizations $\theta = \sigma$ (standard deviation), $\theta = \sigma^2$ (variance), $\theta = \sigma^{-2}$

[1] A function $f : X \to X$ is called equivariant under a transformation $T : X \to X$ if

$$f(T(x)) = T(f(x)).$$

In contrast, a function $f : X \to X$ is called invariant under a transformation $T : X \to X$ if

$$f(T(x)) = f(x).$$

(precision) and $\theta = \log(\sigma)$. The results for these four choices are summarized in Table 2.2.

Parameter	$\frac{\partial^2 \log f(x\|\theta)}{\partial\theta^2}$	$I(\theta)$	Jeffreys prior	$\frac{d\sigma}{d\theta}$
$\theta = \sigma$	$\frac{\theta^2-3x^2}{\theta^4}$	$\frac{2}{\theta^2}$	$\propto \frac{1}{\theta} = \frac{1}{\sigma}$	1
$\theta = \sigma^2$	$\frac{\theta-2x^2}{2\theta^3}$	$\frac{1}{2\theta^2}$	$\propto \frac{1}{\theta} = \frac{1}{\sigma^2}$	$\frac{1}{2\sqrt{\theta}}$
$\theta = \sigma^{-2}$	$-\frac{1}{2\theta^2}$	$\frac{1}{2\theta^2}$	$\propto \frac{1}{\theta} = \sigma^2$	$-\frac{1}{2\theta^{3/2}}$
$\theta = \log(\sigma)$	$-2e^{-2\theta}x^2$	2	$\propto 1$	e^{θ}

Table 2.2: Fisher information and Jeffreys prior for different parametrizations of $\mathcal{N}(0, \sigma^2)$

If σ has the density $\frac{1}{\sigma}$, then by the rules of calculus, $\theta = \theta(\sigma)$ has the density $\frac{1}{\sigma(\theta)}|\frac{d\sigma}{d\theta}|$. So for instance $\theta = \sigma^{-2}$ has the density $\sqrt{\theta}\frac{1}{2\theta^{3/2}} = \frac{1}{2\theta} = \frac{\sigma^2}{2}$ which shows the equivariance. The other cases can be checked similarly.

The argument in the general multiparameter case $\tau = g(\theta)$, where g is one-to-one and differentiable, goes as follows. Denoting the Jacobi matrices of g^{-1} by Dg^{-1}, respectively, the chain rule implies

$$\frac{\partial}{\partial\tau}\log f(x \mid g^{-1}(\tau)) = (Dg^{-1}(\tau))^T \frac{\partial}{\partial\theta}\log f(x \mid g^{-1}(\tau)).$$

Hence, the Fisher information with respect to τ is

$$I_\tau(\tau) = (Dg^{-1}(\tau))^T I_\theta(g^{-1}(\tau)) Dg^{-1}(\tau).$$

It follows that Jeffreys prior for τ is proportional to

$$\det(I_\tau(\tau))^{1/2} = |\det Dg^{-1}(\tau)| \det(I_\theta(g^{-1}(\tau)))^{1/2}.$$

We obtain the same result when applying the change-of-variables formula to

$$\pi(\theta) \propto \det(I_\theta(\theta))^{1/2}.$$

For a scalar parameter, Jeffreys prior is usually a good choice, although it violates the likelihood principle because the Fisher information contains an integral over X. However, for vector parameters, it can have undesirable features as the following example illustrates.

Example 2.4. Vector normal means: Consider the model where X_i $1 \le i \le 2n$ are independent and normally distributed with variance σ^2 and means $\mathbb{E}(X_{2k-1}) = \mathbb{E}(X_{2k}) = \mu_k$. Jeffreys prior for this model is $\pi(\mu_1, \ldots, \mu_n, \sigma^2) \propto \sigma^{-n-2}$, and the posterior is proportional

$$(\sigma^2)^{-n-1}\exp\left(-\frac{1}{4\sigma^2}\sum_{k=1}^{n}(x_{2k}-x_{2k-1})^2\right)\cdot\sigma^{-n}\exp\left(-\frac{1}{\sigma^2}\sum_{k=1}^{n}\left(\mu_k-\frac{x_{2k}+x_{2k-1}}{2}\right)^2\right)$$

The second factor integrated with respect to $\mu_1, \ldots, \mu_n$ gives a constant that does not depend on σ^2. Therefore the first factor is proportional to the marginal posterior of σ^2, and one finds that

$$\mathbb{E}(\sigma^2 \mid x_1, \ldots, x_{2n}) = \frac{1}{4(n-1)}\sum_{k=1}^{n}(x_{2k}-x_{2k-1})^2.$$

As $n \to \infty$, this converges to $\frac{\sigma^2}{2}$. In other words, Jeffreys prior leads to a Bayes estimator with unsatisfactory frequentist properties.

2.2.3 Reference priors

The concept of reference priors was introduced by Bernardo (J. Roy. Statist. Soc. 41, 1979) and it has been studied by Berger and Bernardo in a series of papers afterwards. It has two aspects: A new justification of Jeffreys prior and a distinction between parameters of interest and nuisance parameters.

We begin with the former and call a prior π non-informative if the difference between the prior π and the posterior $\pi(\theta \mid x)$ is maximized in some distance. This seems reasonable because if the data x have the largest possible impact, the impact of the prior is minimal. In the opposite case, the prior is most informative if it is a point mass at some value θ_0 and then the prior and the posterior coincide. There are however two problems with this idea, the choice of the distance and the dependence on the data x. Bernardo proposed to use Kullback-Leibler divergence as the distance measure and to integrate over the data according to the prior predictive distribution $f(x) = \int_\Theta f(x \mid \theta)\pi(\theta)d\theta$. The Kullback-Leibler divergence between two densities f and g is defined as

$$KL(f,g) = \int f(x) \log \frac{f(x)}{g(x)} dx.$$

In general $KL(f,g) \neq KL(g,f)$, it is not a true distance, but it satisfies $KL(f,g) \geq 0$ and $KL(f,g) = 0$ iff $f(x) = g(x)$ for almost all x. Bernardos idea is to maximize

$$\begin{aligned} I(X,\theta) &= \int_X f(x) \int_\Theta \pi(\theta \mid x) \log \frac{\pi(\theta \mid x)}{\pi(\theta)} d\theta dx \\ &= \int_\Theta \pi(\theta) \int_X f(x \mid \theta) \log \frac{\pi(\theta) f(x \mid \theta)}{\pi(\theta) f(x)} dx d\theta \\ &= \int_\Theta \pi(\theta) \int_X f(x \mid \theta) \log \pi(\theta \mid x) dx d\theta - \int_\Theta \pi(\theta) \log \pi(\theta) d\theta. \end{aligned}$$

In information theory $I(X,\theta)$ is called the mutual information of X and θ, confirming the idea behind this approach: A prior is most non-informative if the mutual information of X and θ is maximal.

However, maximizing $I(X,\theta)$ is unfeasible due to the following reasons. First, finding the maximizer is complicated and there is no closed form solution because the marginal $f(x)$ also depends on π. Second and more importantly, the resulting distribution $\pi(\theta)$ typically has a finite support which is a very undesirable property for a prior that is thought to be non-informative. For these reasons, we take the following approach instead. We assume that we have n i.i.d. observations with density $f(x \mid \theta)$ and consider the mutual information $I((X_1, \ldots, X_n), \theta)$ for these n observation. We then standardize $I((X_1, \ldots, X_n), \theta)$, let n go to infinity and take the distribution $\pi(\theta)$ that maximizes this limit as the reference prior.

The limit of $I((X_1, \ldots, X_n), \theta)$ as n goes to infinity is determined as follows. As stated in Section 1.2.4, the posterior $\pi(\theta \mid x)$ does not depend on the prior in the limit and it holds

$$2 \log \pi(\theta \mid x_1, \ldots, x_n) \approx -p \log(2\pi) + \log \det(nI(\widehat{\theta})) - n(\widehat{\theta} - \theta)^T I(\widehat{\theta})(\widehat{\theta} - \theta)$$

where $\widehat{\theta}$ is the MLE and p is the dimension of θ. By the result on the (frequentist) asymptotics of the MLE from the same subsection, $\widehat{\theta}$ is approximately $\mathcal{N}(\theta, \frac{1}{n} I(\theta)^{-1})$. Hence

$$\int \log \det(nI(\widehat{\theta})) \prod_{i=1}^n f(x_i \mid \theta) dx_i \approx p \log n + \log \det I(\theta)$$

and

$$n \int (\widehat{\theta} - \theta)^T I(\widehat{\theta})(\widehat{\theta} - \theta) \prod_{i=1}^{n} f(x_i \mid \theta) dx_i \approx p.$$

See Clark and Barron (1990, IEEE Transactions on Information Theory) for more details. Therefore, we obtain

$$\int \log \pi(\theta \mid x_1, \ldots, x_n) \prod_{i=1}^{n} f(x_i \mid \theta) dx_i \approx \frac{p(\log n - \log(2\pi) - 1) + \log \det I(\theta)}{2}.$$

This finally gives the following approximation for the standardized mutual information

$$I((X_1, \ldots, X_n), \theta) - \frac{p(\log n - \log(2\pi) - 1)}{2} \approx \int_{\Theta} \pi(\theta) \log \frac{\det I(\theta)^{1/2}}{\pi(\theta)} d\theta.$$

The integral on the right hand side is maximal for $\pi(\theta) = c^{-1} \det I(\theta)^{1/2}$ because by concavity of the log it follows that $\log a \leq \log b + (a - b)/b$ with equality iff $a = b$. We thus have again obtained Jeffreys prior in the limit.

Next, we discuss briefly Bernardo's approach if $\theta = (\theta_1, \theta_2)$ can be decomposed in the parameter of interest θ_1 is and the nuisance parameter θ_2. In this case, the prior is factorized as $\pi(\theta) = \pi(\theta_2 \mid \theta_1)\pi(\theta_1)$ and one $\pi(\theta_2 \mid \theta_1)$ as the Jeffreys prior for the model with fixed value θ_1 and $\pi(\theta_1)$ as the Jeffreys prior for the model

$$f^*(x \mid \theta_1) = \int_{\Theta_2} f(x \mid \theta) \pi(\theta_2 \mid \theta_1) d\theta_2.$$

If $\pi(\theta_2 \mid \theta_1)$ is not proper, $f^*(x \mid \theta)$ is not a probability density and one needs to restrict θ_2 first to a compact subset. In the end, one then considers the limit of the prior $\pi(\theta_1)$ as this compact subset tends to the whole space.

In the example 2.4 with normal vector means, we saw that Jeffreys prior leads to an undesirable estimate of σ^2. To correct this, we consider σ^2 as the parameter of interest θ_1 and $\mu_1, \ldots, \mu_n$ as the nuisance parameter θ_2. Then $\pi(\mu_1, \ldots, \mu_n \mid \sigma^2)$ is flat, and

$$f^*(x_1, \ldots, x_{2n} \mid \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left(-\frac{1}{4\sigma^2} \sum_{k=1}^{n} (x_{2k} - x_{2k-1})^2 \right).$$

Although this "density" has infinite total mass, a formal calculation gives $\pi(\sigma^2) \propto 1/\sigma^2$. Finally, one can check that the prior $\pi(\mu_1, \ldots, \mu_n, \sigma^2) \propto 1/\sigma^2$ gives

$$\mathbb{E}(\sigma^2 \mid x_1, \ldots, x_{2n}) = \frac{1}{2(n-2)} \sum_{k=1}^{n} (x_{2k} - x_{2k-1})^2$$

which is much more reasonable than what we had with Jeffreys prior.

2.3 Expert priors

If one takes the view seriously that the prior quantifies knowledge and uncertainty prior to seeing the data, then it makes sense to try to elicit a prior from one or several experts. How to do this best has become a research topic in itself which is situated at the intersection of statistics and psychology. One of the conclusions of research in this area is that expert judgement is subject to various kinds of heuristics and biases, and that the size of unwanted

effects depends strongly on how questions are phrased. A standard procedure in the case of a univariate prior is to first elicit a number of summary statistics. In a second step one then fits a distribution which takes these summaries into account. Taking as summary statistics the median and the quartiles or the 33% and 67% quantiles seems to be a good choice: Expectation, standard deviation and extreme quantiles are more difficult to elicit. Eliciting dependence in multivariate priors is considerably more difficult.

For a more detailed discussion of this issue, see e.g. Garthwaite, Kadane and O'Hagan, J. Amer. Statist. Assoc. 100 (2005), 680-700.

2.4 Discussion

The justification of a chosen prior remains a difficult issue. The concept of a non-informative prior is difficult to implement in complex models with many parameters. Therefore, some subjective choices are often unavoidable. However, in many cases any reasonable choice leads to similar conclusions because the likelihood tends to dominate – at least if the number of observations is large (compare section 1.3.4). In any practical application, one should check that at least marginally the prior is approximately constant in a highest probability density credible set, or do a sensitivity analysis by varying the prior. Choosing a prior such that the desired conclusion is obtained is always possible, but this would be unethical and such a choice would be hard to justify.

The choice of the prior for linear regression models in combination with variable selection – an important practical case – will be discussed in Section 3.2.

If the number of parameters is large compared to the number of observations, then the prior often matters. This seems however unavoidable. In that situation, frequentist statistics often uses regularization methods which usually have a Bayesian interpretation. For instance, if we use penalized maximum likelihood estimation

$$\widehat{\theta} = \arg\max(\log f(x \mid \theta) + P(\theta))$$

the penalty $P(\theta)$ can usually be interpreted as the log of a prior density.

Chapter 3

Hierarchical Bayes models

3.1 Hierarchical models

Until now we distinguished only between parameters θ and observations x. They are both random with a joint distribution that is specified by the marginal for θ and the conditional distribution of x given θ. In hierarchical models, the prior for θ depends on other parameters ξ, called hyperparameters which also have a prior distribution. This leads to a triple of random variables (ξ, θ, x) with joint density

$$\pi(\xi)\pi(\theta \mid \xi)f(x \mid \theta).$$

The basic approach of Bayesian statistics remains unchanged: Once we have specified the three factors of the joint distribution of the triple (ξ, θ, x), we compute the posterior, that is the conditional distribution of the unobserved variables (ξ, θ) given the observed variables by the rules of probability, and then we base our conclusions on this posterior. Often, the primary interest is in the original parameter θ and then we need the marginal posterior $\pi(\theta \mid x)$. There are two ways to compute it.

In the first approach, we begin by computing the marginal prior

$$\pi(\theta) = \int \pi(\theta \mid \xi)\pi(\xi)d\xi$$

and then use Bayes' formula

$$\pi(\theta \mid x) \propto \pi(\theta)f(x \mid \theta).$$

This shows in particular that the introduction of hyperparameters is equivalent to a special choice of a prior for θ.

There are however situations where the approach based on the following formula is computationally easier:

$$\pi(\theta \mid x) = \int \pi(\theta \mid x, \xi)\pi(\xi \mid x)d\xi \propto \int \pi(\theta \mid x, \xi)\pi(\xi)f(x \mid \xi)d\xi, \tag{3.1}$$

where the first equality is obtained by applying the law of total probability to the conditional distribution given x. Further, $\pi(\xi \mid x)$ can be calculated either by marginalizing the joint posterior over θ

$$\pi(\xi \mid x) = \int \pi(\theta, \xi \mid x)d\theta$$

or by using

$$\pi(\xi \mid x) = \frac{\pi(\theta, \xi \mid x)}{\pi(\theta \mid x, \xi)}$$

for any choice of θ. Similarly, $f(x \mid \xi)$ can be obtained by marginalizing the joint distribution of x and θ given ξ

$$f(x \mid \xi) = \int \pi(x, \theta \mid \xi) d\theta = \int f(x \mid \theta) \pi(\theta \mid \xi) d\theta$$

or by using

$$f(x \mid \xi) = \frac{\pi(\theta, x \mid \xi)}{\pi(\theta \mid x, \xi)} = \frac{f(x \mid \theta)\pi(\theta \mid \xi)}{\pi(\theta \mid x, \xi)}.$$

In conjugate situations we have an explicit expression not only for $\pi(\theta \mid x, \xi)$, but usually also for $f(x \mid \xi)$ as we shall show in the examples below. Usually, the final integration over ξ in equation (3.1) cannot be done in closed form, and some approximation is needed. Such approximations will be the topic of Chapter 4.

Example 3.1. Normal means. Let $X_1, \ldots X_n$ be i.i.d. and $\mathcal{N}(\theta, 1)$-distributed, and consider the prior $\theta \sim \mathcal{N}(\mu, \tau^2)$. The posterior conditional on the hyperparameters $\pi(\theta \mid \mu, \tau^2, x)$ is then

$$\theta \mid \mu, \tau^2, x \sim \mathcal{N}\left(\frac{1}{1+n\tau^2}\mu + \frac{n\tau^2}{1+n\tau^2}\bar{x}, \frac{\tau^2}{1+n\tau^2}\right),$$

where $x = (x_1, \ldots, x_n)$. See Example 1.1 for the derivation. This is sensitive to the choice of τ^2, so we can consider τ^2 as hyperparameter to which we assign a distribution. (Of course, the posterior is also sensitive to the choice of μ, but if we want to be non-informative with respect to the prior location, we would take Jeffreys prior $\pi(\mu) \propto 1$). The marginal prior is then a scale mixture of normal priors which is often heavy-tailed. In particular, if we choose a Gamma(γ, λ) prior for τ^{-2}, then the marginal prior is

$$\begin{aligned}
\pi(\theta) &\propto \int (\tau^{-2})^{1/2} \exp\left(-\tau^{-2}\frac{(\theta-\mu)^2}{2}\right) (\tau^{-2})^{\gamma-1} \exp(-\lambda\tau^{-2}) d(\tau^{-2}) \\
&= \int u^{\gamma-1/2} \exp\left(-\left(\lambda + \frac{(\theta-\mu)^2}{2}\right)u\right) du \\
&\propto \frac{1}{\left(\lambda + \frac{(\theta-\mu)^2}{2}\right)^{\gamma+1/2}},
\end{aligned}$$

where the last step follows by a change of the integration variable. This is a scaled and shifted t-density with 2γ degrees of freedom which has heavier tails than the normal distribution. The posterior does not belong to any standard family of distributions, but one can show that the posterior mode

$$\arg\max_\theta -\left(n(\theta-\bar{x})^2 + (2\gamma+1)\log\left(\lambda + \frac{(\theta-\mu)^2}{2}\right)\right)$$

is close to $\bar{x}$ if prior and data are in conflict, that is, if $|\bar{x} - \mu|$ is large.

For the second approach, we already know $\pi(\theta \mid \tau^2, x_1, \ldots, x_n)$. In Example 1.2 we also have found that for given τ^2, the marginal distribution of $(X_1, \ldots, X_n)$ is normal with mean $\mu\mathbf{1}$ and covariance matrix $\Sigma = I + \tau^2\mathbf{1}\mathbf{1}^T$ where $\mathbf{1} = (1, \ldots, 1)^T$. To simplify how $f(x_1, \ldots, x_n \mid \tau^2)$ depends on τ^2, we consider a linear transformation $Y = AX$ where

A is orthogonal and the first row is equal to $\frac{1}{\sqrt{n}}\mathbf{1}^T$. Then the Y_i are independent and the $Y_2, \ldots, Y_n$ are standard normal for any value of μ and τ^2. Hence we obtain that

$$f(x_1, \ldots, x_n \mid \tau^2) \propto (1 + n\tau^2)^{-1/2} \exp\left(-\frac{n}{2(1+n\tau^2)}(\bar{x} - \mu)^2\right)$$

where $\propto$ means up to factors which do not depend on τ^2. This shows that the marginal posterior of τ^2 has little mass for small values, i.e., it favors large values of τ^2, if $(\bar{x}-\mu)^2$ is large which means that the prior mean and data are in conflict.

Example 3.2. Hierarchical Poisson model. In insurance, the number of claims X_j of contract j during a given time period can be modeled as independent Poisson(θ_j) ($j = 1, 2, \ldots, J$) where the θ_j are i.i.d. and Gamma(γ, λ)-distributed (assuming the contracts have similar volumes). On the highest level, we have the parameters (γ, λ), on the second level $(\theta_1, \ldots, \theta_J)$ and on the lowest level $(X_1, \ldots, X_J)$. The joint distribution factors as

$$\pi(\gamma, \lambda) \prod_{j=1}^{J} \pi(\theta_j \mid \gamma, \lambda) f(x_j \mid \theta_j).$$

In the first approach, we would directly work with the prior

$$\pi(\theta_1, \ldots, \theta_J) = \int \frac{\lambda^{J\gamma}}{\Gamma(\gamma)^J} \prod_{j=1}^{J} \theta_j^{\gamma-1} \exp(-\lambda \sum \theta_j) \pi(\lambda, \gamma) d\lambda d\gamma.$$

Again this is not a standard distribution, and in particular the θ_j are not independent under this prior. They are only exchangeable, meaning that the density does not change if we permute the arguments. The second approach is somewhat simpler. First we observe that conditional on (γ, λ), the (θ_j, x_j) are independent for different j's and thus

$$\pi(\theta_1, \ldots, \theta_J \mid \gamma, \lambda, x_1, \ldots, x_J) = \prod_{j=1}^{J} \pi(\theta_j \mid \gamma, \lambda, x_j)$$

and

$$f(x_1, \ldots, x_J \mid \gamma, \lambda) = \prod_{j=1}^{J} f(x_j \mid \gamma, \lambda).$$

Moreover, by conjugacy

$$\pi(\theta_j \mid \gamma, \lambda, x_j) = \frac{(\lambda+1)^{\gamma+x_j}}{\Gamma(\gamma + x_j)} \theta_j^{\gamma+x_j-1} \exp(-(\lambda+1)\theta_j).$$

Because we know the normalizing constant for the Gamma density, we also obtain

$$f(x_j \mid \gamma, \lambda) = \int e^{-\theta_j} \frac{\theta_j^{x_j}}{x_j!} \theta_j^{\gamma-1} e^{-\lambda\theta_j} d\theta_j \frac{\lambda^\gamma}{\Gamma(\gamma)} = \frac{\Gamma(\gamma + x_j)}{x_j! \Gamma(\gamma)} \frac{\lambda^\gamma}{(\lambda+1)^{\gamma+x_j}}.$$

Finally, because $\Gamma(z+1) = z\Gamma(z)$, we end up with

$$f(x_j \mid \gamma, \lambda) = \frac{(\gamma + x_j - 1) \cdots \gamma}{x_j \cdots 1} \left(\frac{\lambda}{\lambda+1}\right)^\gamma \left(\frac{1}{\lambda+1}\right)^{x_j}.$$

This is the negative binomial distribution with parameters γ and $p = \lambda/(\lambda+1)$. For integer γ it is the distribution of the number of failures until γ successes have been

observed in a coin tossing experiment with success parameter p. These results can be used to compute

$$\begin{aligned}\pi(\theta_j \mid x_1, \ldots, x_n) &= \int \pi(\theta_j \mid \gamma, \lambda, x_j)\pi(\gamma, \lambda \mid x_1, \ldots, x_J)d\gamma d\lambda \\ &\propto \int \pi(\theta_j \mid \gamma, \lambda, x_j) \prod_{i=1}^{J} f(x_i \mid \gamma, \lambda)\pi(\gamma, \lambda)d\gamma d\lambda.\end{aligned}$$

The last integral cannot be computed in closed form, but because it is only two-dimensional, approximations are possible. A simpler approach avoiding the computation of an integral will be discussed below in the section on empirical Bayes methods. The formula above implies that observations $x_i \neq x_j$ carry information about θ_j which seems somewhat counterintuitive: The insurance company uses the number of claims of other people to estimate the expected number of claims that I will have in a future period. This is justified by the assumption that I am just one member of a population and the company can estimate the distribution of the expected number of claims in this population from the number of claims of other members of this population.

Example 3.3. The one-way ANOVA model. This model considers outcomes y_{ij} for different groups, indexed by i, each group consisting of a number of subjects, indexed by j:

$$y_{ij} = \theta_i + \varepsilon_{ij} \quad (j = 1, \ldots, n_i; i = 1, \ldots, I),$$

where the ε_{ij} are i.i.d $\sim \mathcal{N}(0, \sigma_\varepsilon^2)$. In order to be consistent with the notation in ANOVA models, we denote here the observations by y and not by x as in the previous examples. In the frequentist approach, we can interpret the group effects θ_i either as unobserved random variables (random effects) or as unknown parameters (fixed effects). The distinction would affect the way that the data are analyzed. In the Bayesian approach, we specify in both interpretations a distribution for $(\theta_1, \ldots, \theta_I)$. In this example one commonly assumes that the θ_i are i.i.d $\sim \mathcal{N}(\mu, \tau^2)$. In ANOVA terminology, μ is the grand mean and $\theta_i - \mu$ is the effect of group i. The hyperparameters are $\xi = (\mu, \tau^2)$, and we let μ and τ^2 be independent a priori. In order to simplify the formulae, we assume σ_ε^2 to be known. The joint distribution of all unknown variables factors as

$$\pi(\mu)\pi(\tau^2) \prod_{i=1}^{I} \pi(\theta_i \mid \mu, \tau^2) \prod_{j=1}^{n_i} f(y_{ij} \mid \theta_i).$$

The priors for μ and τ^2 will be chosen later, the other factors are normal densities.

We will proceed according to the second approach. First we observe that

$$\prod_{j=1}^{n_i} f(y_{ij} \mid \theta_i) = f(\bar{y}_{i\cdot} \mid \theta_i) f(y_{i1}, \ldots, y_{i,n_i-1} \mid \bar{y}_{i\cdot}),$$

where $\bar{y}_{i\cdot} = \sum_{j=1}^{n_i} y_{ij}/n_i$ denotes the mean in group i. That the second factor on the right does not depend on θ_i follows by a similar argument to the one given in Example 3.1. As a consequence we only need to consider the first factor for computing posterior distributions.

By an argument that should be familiar by now we can show that conditional on (y, μ, τ^2) the θ_i are independent and

$$\theta_i \mid y, \mu, \tau^2 \sim \mathcal{N}\left(\frac{n_i\tau^2\bar{y}_{i\cdot} + \sigma_\varepsilon^2\mu}{n_i\tau^2 + \sigma_\varepsilon^2}, \frac{\sigma_\varepsilon^2\tau^2}{n_i\tau^2 + \sigma_\varepsilon^2}\right).$$

In order to compute the posterior of the hyperparameters, $\pi(\mu, \tau^2 \mid y)$, we use

$$\pi(\mu, \tau^2 \mid y) = \pi(\mu, \tau^2 \mid (\bar{y}_{i\cdot})) \propto \pi(\mu)\pi(\tau^2) \prod_{i=1}^{I} f(\bar{y}_{i\cdot} \mid \mu, \tau^2).$$

Because $\bar{y}_{i\cdot} = \theta_i + \bar{\varepsilon}_{i\cdot}$, it follows that conditionally on μ and τ^2, $\bar{y}_{i\cdot}$ is normal with mean μ and variance $\tau^2 + \sigma_\varepsilon^2/n_i$. We could now calculate the marginal posterior $\pi(\theta_i \mid y)$. This leads to a two-dimensional integral with no closed form solution.

Calculations can be somewhat simplified by noting that

$$\pi(\mu, \tau^2 \mid y) = \pi(\mu \mid \tau^2, y)\pi(\tau^2 \mid y) = \pi(\mu \mid \tau^2, (\bar{y}_{i\cdot}))\pi(\tau^2 \mid (\bar{y}_{i\cdot}))$$

and

$$\pi(\mu \mid \tau^2, (\bar{y}_{i\cdot})) \propto \pi(\mu) \prod_{i=1}^{I} f(\bar{y}_{i\cdot} \mid \mu, \tau^2).$$

If we choose the improper flat prior $\pi(\mu) = 1$, we obtain, again by a standard argument, the proper posterior

$$\mu \mid \tau^2, (\bar{y}_{i\cdot}) \sim \mathcal{N}(\widehat{\mu}, V_\mu), \quad \widehat{\mu} = \frac{\sum_{i=1}^{I} w_i \bar{y}_{i\cdot}}{\sum_{i=1}^{I} w_i}, V_\mu = \frac{1}{\sum_{i=1}^{I} w_i}, w_i = \frac{n_i}{n_i\tau^2 + \sigma_\varepsilon^2}. \tag{3.2}$$

I.e., the mean is a convex combination of the group means $\bar{y}_{1\cdot}, \ldots, \bar{y}_{I\cdot}$. Finally, we need to compute $\pi(\tau^2 \mid (\bar{y}_{i\cdot}))$. For this we can either integrate μ out:

$$\pi(\tau^2 \mid (\bar{y}_{i\cdot})) \propto \int \pi(\mu)\pi(\tau^2) \prod_{i=1}^{I} f(\bar{y}_{i\cdot} \mid \mu, \tau^2) d\mu$$

or use the identity

$$\pi(\tau^2 \mid (\bar{y}_{i\cdot})) = \frac{\pi(\mu, \tau^2 \mid (\bar{y}_{i\cdot}))}{\pi(\mu \mid \tau^2, (\bar{y}_{i\cdot}))} \propto \frac{\pi(\mu)\pi(\tau^2) \prod_{i=1}^{I} f(\bar{y}_{i\cdot} \mid \mu, \tau^2)}{\pi(\mu \mid \tau^2, (\bar{y}_{i\cdot}))}$$

which holds for an arbitrary value of μ. If we choose $\mu = \widehat{\mu}$, we obtain

$$\pi(\tau^2 \mid (\bar{y}_{i\cdot})) \propto \pi(\tau^2) V_\mu^{1/2} \prod_{i=1}^{I} \left(\frac{n_i}{\sigma_\varepsilon^2 + n_i\tau^2} \right)^{1/2} \exp\left(-\frac{n_i}{2(\sigma_\varepsilon^2 + n_i\tau^2)} (\bar{y}_{i\cdot} - \widehat{\mu})^2 \right).$$

This is not a standard distribution. As $\tau^2 \to 0$, the terms on the right without the prior $\pi(\tau^2)$ have a finite, strictly positive limit. This implies that for a proper posterior we should choose a prior for τ^2 which is integrable at the origin. This rules out Jeffreys prior, which means that we need some prior information about small values of τ^2.

The main interest is in the posterior distribution of θ_i given y which is given by the formula

$$\pi(\theta_i \mid y) = \int \int \pi(\theta_i \mid y, \mu, \tau^2)\pi(\mu \mid y, \tau^2) d\mu \; \pi(\tau^2 \mid y) d\tau^2.$$

The inner integral can be computed in closed form. It is obtained as the normalizing constant of a normal density with mean equal to a convex combination of all group means $\bar{y}_{1\cdot}, \ldots, \bar{y}_{I\cdot}$. The weights in the convex combination and the variance depend on τ^2, and the integration over τ^2 has to be done by numerical integration or using one of the approximate methods that we will discuss in Chapter 4.

3.2 Empirical Bayes methods

In the examples of the preceding section we have seen that in a hierarchical Bayes model with conjugate priors, we can compute $\pi(\theta \mid x, \xi)$ and $f(x \mid \xi)$ in closed form, but then we need to use approximations to compute

$$\pi(\theta \mid x) \propto \int \pi(\theta \mid x, \xi) f(x \mid \xi) \pi(\xi) d\xi.$$

The empirical Bayes method uses instead

$$\pi(\theta \mid x) \approx \pi(\theta \mid x, \widehat{\xi}(x)), \quad \widehat{\xi}(x) = \arg\max_{\xi} f(x \mid \xi).$$

This means instead of taking a weighted average, we take the value with maximal weight (assuming $\pi(\xi)$ is flat around $\widehat{\xi}(x)$). $\widehat{\xi}(x)$ is simply the marginal maximum likelihood estimator of the hyperparameter.

This method avoids not only the computation of the integral, but also the choice of a hyperprior $\pi(\xi)$. From a conceptual point of view, it is however not quite satisfactory because it uses the data x twice: First to select the prior $\pi(\theta \mid \widehat{\xi}(x))$ and then to compute the posterior according to Bayes' formula. It is also clear that in general the uncertainty is underestimated if we choose any particular value of ξ instead of averaging over different plausible values. For these reasons, Bayesians try to avoid empirical Bayes methods as far as possible, and in fact empirical Bayes methods are usually justified by their frequentist properties. But from a pragmatic point of view, empirical Bayes methods are useful and they offer a way to exit from the infinite hierarchy that arises if we acknowledge that the hyperprior $\pi(\xi)$ is also uncertain and thus we should introduce another prior to describe it, etc.

Let us look what empirical Bayes estimation gives for the examples of the previous section.

Example 3.4. Normal means, ctd. In example 3.1, we have already computed $f(x_1, \ldots, x_n \mid \tau^2)$. Maximizing the result with respect to τ^2 is straightforward. We obtain

$$\widehat{\tau}^2 = \max(0, (\bar{x} - \mu)^2 - 1/n).$$

This means that if the prior mean μ conflicts with the data, we choose a wider prior. In particular, we have

$$\mathbb{E}(\theta \mid x_1, \ldots, x_n, \widehat{\tau}^2) = \begin{cases} \mu & \text{if } |\bar{x} - \mu| \le 1/\sqrt{n}, \\ \bar{x} + \frac{1}{n(\mu - \bar{x})} & \text{if } |\bar{x} - \mu| \ge 1/\sqrt{n}, \end{cases}$$

which converges to θ as $n \to \infty$ since $\bar{x} \to \theta$. However, the posterior variance is zero for $|\bar{x} - \mu| \le 1/\sqrt{n}$ which is not reasonable, confirming the intuition that empirical Bayes underestimates the uncertainty.

Example 3.5. Hierarchical Poisson model, ctd. Maximizing the marginal is equal to maximizing

$$J\gamma \log\left(\frac{\lambda}{1+\lambda}\right) - \sum_{j=1}^{J} x_j \log(1+\lambda) + \sum_{j; x_j > 0} \sum_{k=0}^{x_j - 1} \log(\gamma + k)$$

with respect to γ and λ. Setting the partial derivative with respect to λ equal to zero gives

$$\frac{\widehat{\gamma}}{\widehat{\lambda}} = \frac{1}{J}\sum_{j=1}^{J} x_j = \bar{x},$$

but setting the partial derivative with respect to γ equal to zero gives an equation that does not have an explicit solution. If one does not want to find the minimum numerically, one can use instead of the marginal MLE the marginal moment estimator. By standard rules from probability theory and results about the moments of the Gamma distribution, we obtain

$$\mathbb{E}(X_j \mid \gamma, \lambda) = \mathbb{E}(\mathbb{E}(X_j \mid \theta_j) \mid \gamma, \lambda) = \mathbb{E}(\theta_j \mid \gamma, \lambda) = \frac{\gamma}{\lambda}$$

and

$$\mathrm{Var}(X_j \mid \gamma, \lambda) = \mathrm{Var}(\mathbb{E}(X_j \mid \theta_j) \mid \gamma, \lambda) + \mathbb{E}(\mathrm{Var}(X_j \mid \theta_j) \mid \gamma, \lambda) = \frac{\gamma}{\lambda^2} + \frac{\gamma}{\lambda} = \frac{\gamma(\lambda+1)}{\lambda^2}.$$

(alternatively, we could use known results for the moments of the negative binomial distribution). Therefore the marginal moment estimator is

$$\widehat{\lambda} = \frac{\bar{x}}{S_J^2 - \bar{x}}, \quad \widehat{\gamma} = \bar{x}\widehat{\lambda}$$

where S_J^2 is the sample variance $(J-1)^{-1}\sum_j (x_j - \bar{x})^2$. The empirical Bayes estimate of θ_j is therefore

$$\mathbb{E}(\theta_j \mid x_j, \widehat{\lambda}, \widehat{\gamma}) = \frac{\widehat{\lambda}}{\widehat{\lambda}+1}\bar{x} + \frac{1}{\widehat{\lambda}+1}x_j.$$

It shrinks the individual experience x_j towards the average experience of all contracts. Because $\widehat{\lambda}$ is a decreasing function of S_J^2, the shrinkage is less if the portfolio of all contracts is heterogeneous. The method breaks down if $S_J^2 \le \bar{x}$.

Example 3.6. The one-way ANOVA model, ctd. In order to simplify the formulae, we assume that $\sigma_\varepsilon^2 = 1$ and $n_i = 1$ for all i. Then we need only a single index i. Because conditionally on μ and τ^2, $y_i = \theta_i + \varepsilon_i$ is normal with mean μ and variance $\tau^2 + 1$, the marginal MLE is easily seen to be

$$\widehat{\mu} = \bar{y}, \quad \widehat{\tau}^2 = \max\left(\frac{1}{I}\sum_{i=1}^{I}(y_i - \bar{y})^2 - 1, 0\right).$$

Therefore, for $\widehat{\tau}^2 > 0$, the empirical Bayes estimator of θ_i is

$$\mathbb{E}(\theta_i \mid y, \widehat{\mu}, \widehat{\tau}^2) = \frac{I}{\sum_{i=1}^{I}(y_i - \bar{y})^2}\ \bar{y} + \left(1 - \frac{I}{\sum_{i=1}^{I}(y_i - \bar{y})^2}\right)\ y_i.$$

Again we have a shrinkage of the individual observation towards the mean of all observations, and the amount of shrinkage depends on how heterogeneous the sample is. As in the previous example, there are problems when $\sum_{i=1}^{I}(y_i - \bar{y})^2 < 1$ because y_i has then a negative weight.

A famous result of Charles Stein says that if we use the loss function $L(t, \theta) = \sum_i (t_i - \theta_i)^2$ and if $I > 3$, then this empirical Bayes estimator has, for all θ, a smaller frequentist risk than the unbiased estimator $\widehat{\theta}_i = y_i$. This is a frequentist justification for an empirical Bayes method.

3.3 Model selection in linear regression

We consider here the linear regression model

$$y = \alpha\mathbf{1} + X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I).$$

Here y is an $n \times 1$ vector of responses, $\mathbf{1} = (1, \ldots, 1)^T$, X is the $n \times p$ design matrix, α is the intercept, β the $p \times 1$ regression parameter and ε is the $n \times 1$ vector of errors. The j-th column of X contains the values of the j-th explanatory variable, and we assume that all columns are centered: $X^T\mathbf{1} = 0$. We consider not only estimation of the parameters, but also the selection of explanatory variables. Let γ denote an element of $\{0,1\}^p$ where $\gamma_j = 1$ iff the j-th variable is selected. Furthermore, let β_γ be the subvector that contains only the selected components and X_γ the corresponding submatrix. Then the model indexed by γ is

$$y = \alpha\mathbf{1} + X_\gamma\beta_\gamma + \varepsilon.$$

The number of explanatory variables included in the model γ will be denoted by $|\gamma|$. The unknowns are $(\gamma, \beta_\gamma, \alpha, \sigma^2)$ and we would like to compute the posterior of these unknowns. For this we need the likelihood and a prior.

3.3.1 g-prior and posterior for fixed g and γ

The likelihood is

$$(\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}(y - \alpha\mathbf{1} - X_\gamma\beta_\gamma)^T(y - \alpha\mathbf{1} - X_\gamma\beta_\gamma)\right).$$

Because the columns of X are assumed to be centered, the MLE for α and β_γ is

$$\widehat{\alpha} = \bar{y}, \quad \widehat{\beta}_\gamma = (X_\gamma^T X_\gamma)^{-1} X_\gamma^T y,$$

and the likelihood is equal to

$$\begin{aligned} & (\sigma^2)^{-n/2} \exp\left(-\frac{(y - \widehat{\alpha}\mathbf{1} - X_\gamma\widehat{\beta}_\gamma)^T(y - \widehat{\alpha}\mathbf{1} - X_\gamma\widehat{\beta}_\gamma) + n(\widehat{\alpha} - \alpha)^2 + (\beta_\gamma - \widehat{\beta}_\gamma)^T X_\gamma^T X_\gamma(\beta_\gamma - \widehat{\beta}_\gamma)}{2\sigma^2}\right) \\ = \quad & (\sigma^2)^{-n/2} \exp\left(-\frac{s_\gamma^2 + n(\widehat{\alpha} - \alpha)^2 + (\beta_\gamma - \widehat{\beta}_\gamma) X_\gamma^T X_\gamma(\beta_\gamma - \widehat{\beta}_\gamma)}{2\sigma^2}\right) \end{aligned}$$

where

$$s_\gamma^2 = (y - \widehat{\alpha}\mathbf{1} - X_\gamma\widehat{\beta}_\gamma)^T(y - \widehat{\alpha}\mathbf{1} - X_\gamma\widehat{\beta}_\gamma)$$

is the residual sum of squares in the model defined by γ.

In this subsection, we fix the model γ. We assume α to be independent a priori from σ^2 and β_γ, and we take univariate Jeffreys priors for α and σ^2. This means

$$\pi(\beta_\gamma, \alpha, \sigma^2) = \pi(\alpha)\pi(\sigma^2)\pi(\beta_\gamma \mid \sigma^2) \propto \pi(\beta_\gamma \mid \sigma^2)\sigma^{-2}.$$

For the first factor, a popular choice is the so-called g-prior of Zellner

$$\beta_\gamma \mid \sigma^2 \sim \mathcal{N}(\beta_\gamma^0, g\sigma^2(X_\gamma^T X_\gamma)^{-1}).$$

The g-prior is a middle ground between being informative and completely non-informative $\pi(\beta_\gamma \mid \sigma^2) \propto 1$. The idea is to allow for (possibly weak) prior information about β_γ and to

bypass the specification of the most difficult aspect of the prior $\pi(\beta_\gamma \mid \sigma^2)$, i.e., the prior correlation structure of β_γ. For the prior correlation structure determined by $(X_\gamma^T X_\gamma)^{-1}$, we have a similar "non-informativeness" property as for Jeffreys prior: Since for the MLE $\hat{\beta}_\gamma$, $\text{var}(\hat{\beta}_\gamma) = (X_\gamma^T X_\gamma)^{-1}\hat{\sigma}^2$, the prior puts more mass in areas of the parameter space where we expect the data to be more informative about β on average. $g > 0$ is a hyperparameter which can be interpreted as a measure of the amount of information available in the prior relative to the data. Specifically, the above formula for the likelihood shows that the g-prior of Zellner arises as the posterior from a flat prior, a response vector $y = X_\gamma \beta_\gamma^0$ with the same design matrix X_γ, no intercept, and error variance $g\sigma^2$. For g large, this prior is therefore weakly informative, and for $g \to \infty$, we approach a flat prior. But we cannot use a flat prior if we want to calculate Bayes factors and do model selection later because this would leave posterior probabilities of different models γ undefined (see below). The dependence of $\pi(\beta_\gamma \mid \sigma^2)$ on σ^2 is necessary for conjugacy. β_γ^0 is the prior mean. Often, one uses $\beta_\gamma^0 = 0$. Because in regression models the design matrix is considered to be known and fixed, it is allowed to use it for the g-prior.

Combining this prior with the likelihood above leads to the posterior

$$\begin{aligned}
&\pi(\beta_\gamma, \alpha, \sigma^2 \mid y) \\
&\propto (\sigma^2)^{-n/2-1} \exp\left(-\frac{s_\gamma^2}{2\sigma^2}\right) (g\sigma^2)^{-|\gamma|/2} \det(X_\gamma^T X_\gamma)^{1/2} \\
&\quad \cdot \exp\left(-\frac{n(\widehat{\alpha}-\alpha)^2 + \left(\beta_\gamma - \widehat{\beta}_\gamma\right)^T X_\gamma^T X_\gamma \left(\beta_\gamma - \widehat{\beta}_\gamma\right) + \frac{1}{g}\left(\beta_\gamma - \beta_\gamma^0\right)^T X_\gamma^T X_\gamma \left(\beta_\gamma - \beta_\gamma^0\right)}{2\sigma^2}\right).
\end{aligned}$$

As in previous examples involving the normal distribution, we complete the square

$$\begin{aligned}
&\left(\beta_\gamma - \widehat{\beta}_\gamma\right)^T X_\gamma^T X_\gamma \left(\beta_\gamma - \widehat{\beta}_\gamma\right) + \frac{1}{g}\left(\beta_\gamma - \beta_\gamma^0\right)^T X_\gamma^T X_\gamma \left(\beta_\gamma - \beta_\gamma^0\right) \\
&\quad = \frac{g+1}{g}\left(\beta_\gamma - \left(\frac{g}{g+1}\widehat{\beta}_\gamma + \frac{1}{g+1}\beta_\gamma^0\right)\right)^T X_\gamma^T X_\gamma \left(\beta_\gamma - \left(\frac{g}{g+1}\widehat{\beta}_\gamma + \frac{1}{g+1}\beta_\gamma^0\right)\right) \\
&\qquad + \frac{1}{g+1}\left(\widehat{\beta}_\gamma - \beta_\gamma^0\right)^T X_\gamma^T X_\gamma \left(\widehat{\beta}_\gamma - \beta_\gamma^0\right).
\end{aligned}$$

Then the posterior becomes

$$\begin{aligned}
\pi(\beta_\gamma, \alpha, \sigma^2 \mid y) \quad &\propto \quad \left(\frac{n}{\sigma^2}\right)^{1/2} \exp\left(-\frac{n}{2\sigma^2}(\widehat{\alpha}-\alpha)^2\right) \cdot \left(\frac{g+1}{g\sigma^2}\right)^{|\gamma|/2} \det(X_\gamma^T X_\gamma)^{1/2} \\
&\cdot \quad \exp\left(-\frac{g+1}{2g\sigma^2}\left(\beta_\gamma - \left(\frac{g}{g+1}\widehat{\beta}_\gamma + \frac{1}{g+1}\beta_\gamma^0\right)\right)^T X_\gamma^T X_\gamma \left(\beta_\gamma - \left(\frac{g}{g+1}\widehat{\beta}_\gamma + \frac{1}{g+1}\beta_\gamma^0\right)\right)\right) \\
&\cdot \quad (\sigma^2)^{-(n+1)/2}(g+1)^{-|\gamma|/2} \exp\left(-\frac{s_\gamma^2 + \frac{1}{g+1}\left(\widehat{\beta}_\gamma - \beta_\gamma^0\right)^T X_\gamma^T X_\gamma \left(\widehat{\beta}_\gamma - \beta_\gamma^0\right)}{2\sigma^2}\right).
\end{aligned}$$

From this, we conclude that

$$\beta_\gamma \mid y, \sigma^2 \sim \mathcal{N}\left(\frac{g}{g+1}\widehat{\beta}_\gamma + \frac{1}{g+1}\beta_\gamma^0, \frac{g\sigma^2}{g+1}(X_\gamma^T X_\gamma)^{-1}\right)$$

and

$$\alpha \mid y, \sigma^2 \sim N\left(\bar{y}, \frac{\sigma^2}{n}\right),$$

independently of β_γ. In particular, we have

$$\begin{aligned}\mathbb{E}(\beta_\gamma \mid y) &= \mathbb{E}(\mathbb{E}(\beta_\gamma \mid \sigma^2, y) \mid y) \\ &= \mathbb{E}\left(\frac{g}{g+1}\widehat{\beta}_\gamma + \frac{1}{g+1}\beta_\gamma^0 \mid y\right) \\ &= \frac{g}{g+1}\widehat{\beta}_\gamma + \frac{1}{g+1}\beta_\gamma^0.\end{aligned}$$

Furthermore, integrating with respect to α and β_γ gives us

$$\pi(\sigma^2 \mid y) \propto (\sigma^2)^{-n/2-1} \exp\left(-\frac{1}{2\sigma^2}\left(s_\gamma^2 + \frac{1}{g+1}\left(\widehat{\beta}_\gamma - \beta_\gamma^0\right)^T X_\gamma^T X_\gamma \left(\widehat{\beta}_\gamma - \beta_\gamma^0\right)\right)\right).$$

This means that

$$\sigma^{-2} \mid y \sim \text{Gamma}\left(\frac{n-1}{2}, \frac{s_\gamma^2 + \frac{1}{g+1}\left(\widehat{\beta}_\gamma - \beta_\gamma^0\right)^T X_\gamma^T X_\gamma \left(\widehat{\beta}_\gamma - \beta_\gamma^0\right)}{2}\right).$$

3.3.2 Model selection

For Bayesian model selection, we put a prior on the set of possible models γ and compute the posterior probability

$$\pi(\gamma \mid y) = \frac{\pi(\gamma) f(y \mid \gamma)}{\sum_{\gamma'} \pi(\gamma') f(y \mid \gamma')}$$

where the marginal likelihood $f(y \mid \gamma)$ is

$$f(y \mid \gamma) = \int f(y \mid \beta_\gamma, \alpha, \sigma^2) \pi(\beta_\gamma, \alpha, \sigma^2) d\beta_\gamma d\alpha d\sigma^2.$$

If we had chosen an improper prior for β_γ, posterior model probabilities would not be well defined even if the posterior $\pi(\beta_\gamma \mid y)$ is well defined: An improper prior means that $f(y \mid \gamma)$ is only defined up to an arbitrary constant which does not cancel in $\pi(\gamma \mid y)$ because this constant differs for different models. An improper prior for α and σ^2 is allowed because these two parameters are shared by all models.

For the g-prior, $f(y \mid \gamma)$ can be computed in closed form by the same arguments that were used to compute the posterior $\pi(\beta_\gamma, \alpha, \sigma^2 \mid y)$. For the sake of simplicity, we assume $\beta_\gamma^0 = 0$ in the following. One can show that

$$f(y \mid \gamma) \propto \frac{(1+g)^{(n-1-|\gamma|)/2}}{(1+g(1-R_\gamma^2))^{(n-1)/2}}$$

where $\propto$ means up to factors which contain neither γ nor g, where

$$R_\gamma^2 = 1 - \frac{s_\gamma^2}{s_0^2}$$

is the so-called coefficient of determination of model γ, and where $s_0^2 = (y - \bar{y}\mathbf{1})^T(y - \bar{y}\mathbf{1})$ is the sum of squared errors in the null model $\gamma = 0$.

The above result is obtained by using $f(y \mid \beta_\gamma, \alpha, \sigma^2)\pi(\beta_\gamma, \alpha, \sigma^2) \propto \pi(\beta_\gamma, \alpha, \sigma^2 \mid y)$, integrating out over β_γ, α and σ^2, and using the relation

$$s_\gamma^2 + \frac{1}{g+1}\widehat{\beta}_\gamma^T X_\gamma^T X_\gamma \widehat{\beta}_\gamma = \frac{s_0^2(1+g(1-R_\gamma^2))}{g+1}.$$

The last relation follows from

$$\widehat{\beta}_\gamma^T X_\gamma^T X_\gamma \widehat{\beta}_\gamma = s_0^2 - s_\gamma^2$$

which holds true due to the orthogonality properties of least squares.

If we want to use posterior model probabilities, we also have to choose a prior for $\gamma \in \{0,1\}^p$. The simplest choice is the uniform prior $\pi(\gamma) = 2^{-p}$ for all γ which means that every explanatory variable is included with probability $\frac{1}{2}$ independently of the others. For large p, this is however informative for the size of the model because with high prior probability $|\gamma| \approx \frac{p}{2}$. In order to have an uniform prior for $|\gamma|$ one can assume that every explanatory variable is included with probability r where r is unknown and uniform on $(0,1)$. If the number of variables p is large, then computing $\pi(\gamma \mid y)$ for all γ and finding the a posteriori most likely model is difficult. In such a situation, stochastic search algorithms which avoid enumerating the whole space $\{0,1\}^p$ systematically are being recommended.

The fact that the posterior model probabilities $\pi(\gamma \mid y)$ also depend on the prior can be considered a disadvantage. One can avoid this if one does Bayesian model comparison based on the Bayes factor, the ratio of posterior and prior odds for two models, which is independent of the prior:

$$B(\gamma, \gamma') = \frac{\pi(\gamma \mid y)}{\pi(\gamma' \mid y)} \frac{\pi(\gamma')}{\pi(\gamma)} = \frac{f(y \mid \gamma)}{f(y \mid \gamma')} = (1+g)^{(|\gamma'|-|\gamma|)/2} \left(\frac{1+g(1-R_{\gamma'}^2)}{1+g(1-R_\gamma^2)} \right)^{(n-1)/2}. \quad (3.3)$$

In particular, the Bayes factor for comparison with the null model is

$$B(\gamma, 0) = \frac{(1+g)^{(n-|\gamma|-1)/2}}{(1+g(1-R_\gamma^2))^{(n-1)/2}}.$$

The second factor in the right-hand side of (3.3) compares how well the two models fit the data: If γ' is a sub-model of γ, that is every variable included in γ' is also included in γ, then by the definition of least squares $s_{\gamma'}^2 \geq s_\gamma^2$ and $R_{\gamma'}^2 \leq R_\gamma^2$. The second factor is therefore greater (or equal) to one. Further, the first factor is a penalty for the more complex model (measured by the number of parameters). In particular it is less than one if γ' is a sub-model of γ. Hence the Bayes factor balances goodness of fit and complexity when comparing two models.

Clearly, the Bayes factor strongly depends on the chosen value of g. In an attempt to be non informative, one is tempted to let g tend to infinity. However, as $g \to \infty$, $B(\gamma, 0) \to 0$ for any $\gamma \neq 0$, that is we always choose the null model ("Bartlett's paradox"). Choosing any fixed value for g also leads to problems: If one model γ has an excellent fit, one would expect that the Bayes factor clearly favors this model over the null model. However, if g is fixed and $R_\gamma^2 \to 1$ then $B(\gamma, 0) \to (1+g)^{(n-1-|\gamma|)/2}$ which is finite although the evidence in favor of γ as measured, for instance, by the p value of a corresponding F-test goes to infinity ("Information paradox").

For prediction of a new observation y_{n+1} for a given vector x_{n+1} of explanatory variables, a Bayesian prefers to use model averaging instead of model selection. For instance, the

prediction of the mean of y_{n+1} is (for known g) given by

$$\mathbb{E}(y_{n+1} \mid y) = \bar{y} + \frac{g}{g+1} \sum_{\gamma} x_{n+1,\gamma}^T \widehat{\beta}_\gamma \pi(\gamma \mid y).$$

3.3.3 Unknown g

In order to avoid the paradoxes described in the previous section, we should consider g to be unknown. We can then either use an empirical Bayes approach or a fully Bayesian approach with a hyperprior on g.

In the empirical Bayes approach, we can determine $\widehat{g}$ either separately for every model γ, or globally. The former means

$$\begin{aligned} \widehat{g} &= \arg\max \left((n-1-|\gamma|)\log(1+g) - (n-1)\log(1+g(1-R_\gamma^2)) \right) \\ &= \max\left(\frac{(n-1-|\gamma|)R_\gamma^2}{|\gamma|(1-R_\gamma^2)} - 1, 0 \right). \end{aligned}$$

The above ratio is the standard F-test statistics for the null hypothesis $\beta_\gamma = 0$. A large value of this test statistic means that the data are in conflict with the prior mean $\beta_\gamma = 0$ and, therefore, the influence of the prior is reduced. In the latter case, we have

$$\widehat{g} = \arg\max \sum_{\gamma} \pi(\gamma) f(y \mid \gamma)$$

which has to be computed numerically. In both cases, one can show that the information paradox does not occur any more.

When choosing g, an arguably desirable frequentist property is the so-called model selection consistency: the probability that the true model γ' is selected must converge to 1 as the number of samples n goes to infinity:

$$\pi(\gamma' \mid y) \xrightarrow{p} 1 \text{ as } n \to \infty.$$

The above empirical Bayes approach does not have the model selection consistency property when the true model is the null model. Otherwise, the empirical Bayes approach has model selection consistency. In contrast, when using a fully Bayesian approach, we can have the model selection consistency property for all true models.

In the fully Bayesian approach, it is desirable to have a prior $\pi(g)$ such that

$$f(y \mid \gamma) \propto \int \frac{(1+g)^{(n-1-|\gamma|)/2}}{(1+g(1-R_\gamma^2))^{(n-1)/2}} \pi(g) dg$$

can be computed easily. Moreover, in order to avoid the information paradox, it is sufficient to have

$$\int (1+g)^{(n-1-|\gamma|)/2} \pi(g) dg = \infty \quad (|\gamma| \leq p).$$

In the case of the minimal sample size $n = p + 2$, this means $\int (1+g)^{1/2} \pi(g) dg = \infty$. In the literature, the choices

$$\pi(g) \propto g^{-3/2} \exp(-n/(2g))$$

and

$$\pi(g) \propto (1+g)^{-a/2}, \quad (2 < a \leq 3),$$

have been proposed. The former choice together with Zellners g-prior is called the Zellner-Siow prior and the latter is the so-called hyper-g prior. For the hyper-g prior one can express $f(y \mid \gamma)$ with the so-called Gaussian hypergeometric function (Abramowitz and Stegun, 1970, Chapter 15). Moreover, the shrinkage factor $\mathbb{E}(\frac{g}{g+1} \mid y, \gamma)$ which appears in $\mathbb{E}(\beta_\gamma \mid y, \gamma)$ can also be expressed with the same Gaussian hypergeometric function. The Zellner-Siow prior has the model selection consistency property for all true models whereas the hyper-g prior has this property for all true models except for the null model.

For more details on Bayesian model selection in regression, see Liang et al., J. Amer. Statist. Assoc. 103, 2008, p. 410.

3.4 Gaussian processes

In Bayesian non-parametric regression, stochastic processes are used to define a prior on the space of all regression functions. We focus here on Gaussian processes. A Gaussian process is a collection of random variables $\{z(x); x \in D \subset \mathbb{R}^p\}$ for which any finite-dimensional distribution is Gaussian. $x = (x^1, \ldots, x^p)$ is a vector of explanatory variables, also called input locations, and D is the domain of these explanatory variables. Finite-dimensional distributions are joint distributions of $(z(x_1), \ldots, z(x_n))$ for any n and any values $x_1, \ldots, x_n \in D \subset \mathbb{R}^p$. Because Gaussian distributions are determined by their first two moments, a Gaussian process is specified by a mean function

$$\begin{aligned} m\colon D &\longrightarrow \mathbb{R} \\ x &\longmapsto m(x) \end{aligned}$$

and a covariance function

$$\begin{aligned} C\colon D \times D &\longrightarrow \mathbb{R} \\ (x, x') &\longmapsto C(x, x'). \end{aligned}$$

A function is a covariance function if it is symmetric and positive definite:

$$C(x, x') = C(x', x) \quad \forall x, x' \in D$$

and

$$\sum_{i,j=1}^{n} C(x_i, x_j)\beta_i\beta_j \geq 0 \quad \forall x_i \in D, n \in \mathbb{N}, \beta_i \in \mathbb{R}.$$

In time series and in spatial and spatio-temporal statistics, we usually have observations from only one realization of the underlying random function, but there are also situations where multiple realizations are available (e.g. meta-learning). It is clear that from one realization it is not possible to find out something about the underlying finite dimensional distributions without additional assumptions. We need some redundancy for doing statistics. The simplest assumption which provides redundancy is stationarity.

A Gaussian process is called stationary if the finite-dimensional distributions do not change when the points x are shifted by a common vector h, that is $(z(x_1), \ldots, z(x_n))$ and $(z(x_1 + h), \ldots, z(x_n + h))$ have the same distribution

$$(z(x_1), \ldots, z(x_n)) \stackrel{d}{=} (z(x_1 + h), \ldots, z(x_n + h)).$$

It is easy to see that a Gaussian process is stationary if the mean function is constant, $m(x) = m$, and the covariance function depends only on the relative position $C(x, x') =$

$C(x-x')$. Since only the difference $h = x - x'$ matters for stationary covariance functions, we will write $C(h)$ in the following instead of $C(x, x')$.

The simplest form of non-stationarity has a non-constant mean whereas the covariance is still stationary. For instance, one can use a linear function to model the mean

$$\begin{aligned} z(x) &= \beta_0 + \sum_{j=1}^{q} \beta_j f_j(x) + \widetilde{z}(x), \\ &= f(x)^T \beta + \widetilde{z}(x) \end{aligned} \tag{3.4}$$

where $\beta = (\beta_0, \ldots, \beta_q)^T$, $f(x) = (1, f_1(x), \ldots, f_q(x))^T$, $f_1(), \ldots, f_q()$ are some known functions (e.g., longitude, latitude, or altitude), and $\widetilde{z}(x)$ is a stationary Gaussian process with mean zero.

Besides invariance under translations, one can also consider invariance under rotations. A stationary Gaussian process is called isotropic if $C(h) = C(\|h\|)$. For an isotropic process, the dependence is the same in all directions. If z is an isotropic stationary process and T is any invertible $p \times p$ matrix which is not a multiple of the identity, then the process $z'(x) = z(Bx)$ is again stationary but no longer isotropic:

$$\mathbb{E}\big[z'(x)\big] = m, \qquad \mathrm{Cov}\big(z'(x+h), z'(x)\big) = C\big(\|T(x+h) - Tx\|\big) = C\big(\|Th\|\big). \tag{3.5}$$

Therefore, the shape of the covariance function is the same in all directions, but the scaling of the distance depends on the direction.

3.4.1 Covariance functions

As covariance function, one often uses a parametric function $C()$ parameterized by some hyperparameters θ. One of the most widely used class of covariance functions is the Matérn covariance:

$$C(h) = \sigma^2 \frac{1}{2^{\nu-1}\Gamma(\nu)} \big(\sqrt{2\nu}\,\|h\|/\rho\big)^{\nu} K_\nu\big(\sqrt{2\nu}\,\|h\|/\rho\big),$$

where $K_\nu(\cdot)$ is the so-called modified Bessel function of order ν and $\Gamma(\cdot)$ the Gamma function. For this covariance, $\theta = (\sigma^2, \rho, \nu)$. The parameter σ^2 is the marginal variance of $z(s)$. ρ is called range parameter which controls how fast the covariance function decays to zero, i.e., it determines the "amount" of the spatial dependence. Note that σ scales the "output" amplitude ($z \mapsto \text{const} \times z$), and ρ scales the distance between input points ($x \mapsto \text{const} \times x$). Finally, ν is a shape parameter which determines how smooth $C()$ is at the origin. This parameter also determines the smoothness of the corresponding process. More precisely, the corresponding process $z(x)$ is m-times mean-square differentiable[1] if and only if $\nu > m$, and it has almost surely m times continuously differentiable realizations if $\nu > m + p/2$. In Figure 3.1, we show several choices of Matérn covariance functions and samples from corresponding Gaussian processes for different ν's and ρ's.

There are some special cases of Matérn covariance functions. For $\nu = \frac{1}{2}$, we obtain

$$C(h) = \sigma^2 \exp\big(-\|h\|/\rho\big),$$

the so-called the exponential model, and for $\nu \to \infty$, we obtain

$$C(h) = \sigma^2 \exp\big(-\|h\|^2/(2\rho^2)\big),$$

[1] See the appendix for a definition

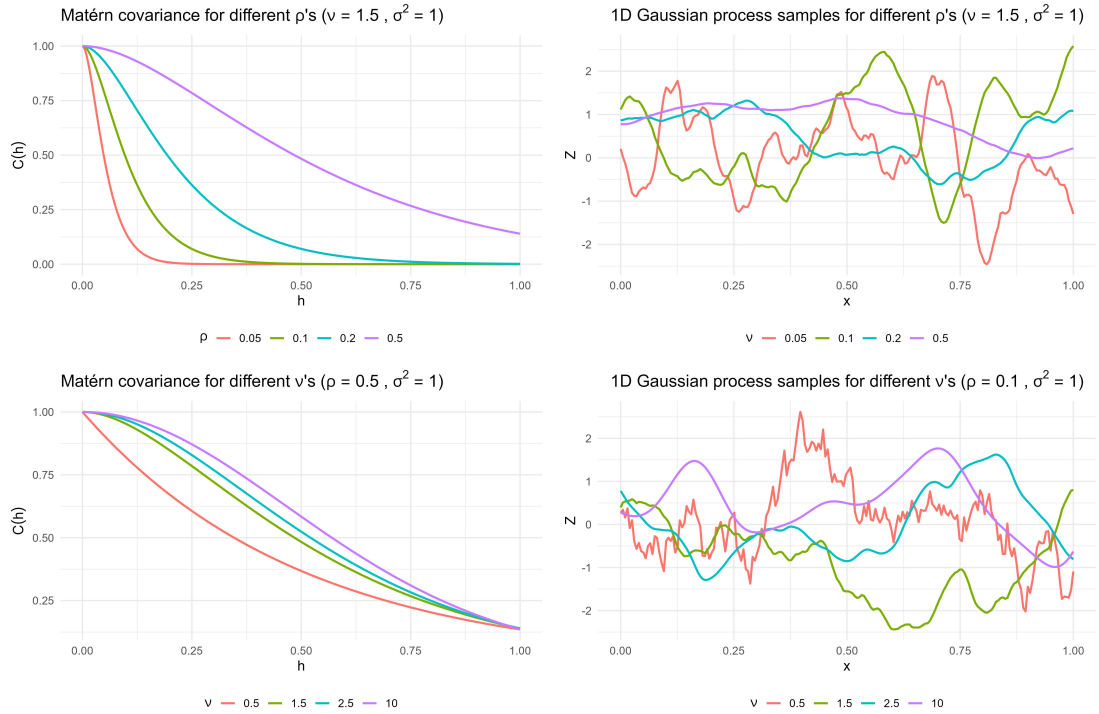


Figure 3.1: Matérn covariance functions and samples from corresponding Gaussian processes for different ν's and ρ's.

which is called a Gaussian covariance function. In general, for half-integer $\nu = m + 0.5$, where m is integer, the Matérn covariance is the product of an exponential covariance and a polynomial of order m.

Note that there are different parametrizations of this Matérn model in the literature. The reason for not absorbing the factor $\sqrt{2\nu}$ in ρ is that models with different values of ν differ mainly with respect to their smoothness near zero, whereas the distance where the covariance is close to zero is similar for all ν, and ρ is thus better comparable and interpretable.

The Matérn covariance introduced above is a stationary and isotropic model. The simplest non-isotropic extension is to use the construction in (3.5) with T being a diagonal matrix:

$$\begin{aligned} C(h) &= \tilde{C}\big(\|Th\|\big), \\ T &= \operatorname{diag}(1/\rho_1, \ldots, 1/\rho_p), \end{aligned}$$

where $\tilde{C}$ is a Matérn covariance function with $\rho = 1$. Every coordinate x^j now has its own range parameter ρ_j. Such covariance functions are frequently used in machine learning where they are often called automatic relevance determination (ARD) covariances since $1/\rho_j$ determines how relevant variable j is.

3.4.2 Estimation

In the following, we assume the Gaussian process prior

$$z(x) = f(x)^T \beta + \widetilde{z}(x)$$

where $\widetilde{z}(x)$ is a mean-zero GP with covariance $C_\theta(\dot{;}\dot{)}$, and that we observe the Gaussian process at n locations x_i with a (measurement) error term:

$$y(x_i) = z(x_i) + \varepsilon(x_i), \quad i = 1, \ldots, n, \quad \varepsilon(x_i) \overset{iid}{\sim} N(0, \sigma_\varepsilon^2).$$

We denote $y = (y(x_1), \ldots, y(x_n))$. The parameters β and θ are hyperparameters. One can either use prior distributions for them or an empirical Bayes approach. We focus on the latter. For notational simplicity, we include σ_ε^2 in θ in the following. For instance, for an isotropic Matérn covariance, $\theta = (\sigma_\varepsilon^2, \sigma^2, \rho, \nu)$. To obtain the marginal likelihood, we note that

$$y \sim N(\mu(\beta), \Sigma(\theta)),$$

where

$$\mu(\beta) = F\beta,$$

F is an $n \times (q+1)$ design matrix whose row i is $f(x_i)$, and $\Sigma(\theta) = \mathrm{Cov}(y)$ is an $n \times n$ matrix with entries

$$\Sigma(\theta)_{ij} = C_\theta(x_i, x_j) + \sigma_\varepsilon^2 1_{i=j}, \quad 1 \le i, j \le n.$$

The log-marginal likelihood

$$\ell(\beta, \theta) = \log(f(y|\beta, \theta)) = \log\left(\int f(y|z, \theta)\pi(z|\beta, \theta)dz\right),$$

where $z = (z(x_1), \ldots, z(x_n))$, is thus given by

$$\ell(\beta, \theta) = -1/2\left(n\log(2\pi) \; + \; \log\det\big(\Sigma(\theta)\big) \; + \; \big(y - \mu(\beta)\big)^\top \Sigma(\theta)^{-1}\big(y - \mu(\beta)\big)\right).$$

This is maximized in an empirical Bayes approach. The parameters $\widehat{\beta}(\theta)$ which maximize the likelihood for a fixed value of θ are the so-called generalized least squares (GLS) estimators:

$$\widehat{\beta}(\theta) = \big(F^\top \Sigma(\theta)^{-1} F\big)^{-1} F^\top \Sigma(\theta)^{-1} y. \tag{3.6}$$

If we plug this into the formula for the likelihood, we obtain the so-called profile likelihood $\ell(\widehat{\beta}(\theta), \theta)$ which is a function of θ alone. Optimizing this profile likelihood can be computationally more efficient as the optimization problem is of lower dimension.

3.4.3 Prediction

Predicting $z^p = (z(x_1^p), \ldots, z(x_{n_p}^p))$ at n_p new locations $(x_1^p, \ldots, x_{n_p}^p)$ is done using the posterior predictive distribution which is given by

$$\begin{aligned} \pi(z^p|y, \beta, \theta) &= N(\mu^p, \Sigma^p), \\ \mu^p &= F^p\beta + \Sigma^{p,o}\Sigma^{-1}\big(y - F\beta\big), \\ \Sigma^p &= \Sigma^{p,p} - \big(\Sigma^{p,o}\big)^T \Sigma^{-1}\Sigma^{p,o}, \end{aligned} \tag{3.7}$$

where F^p is the design matrix at the prediction locations, $\Sigma = \mathrm{Cov}(y)$, $\Sigma^{p,o}$ is a $n_p \times n$ cross-covariance matrix with entries

$$\Sigma_{ij}^{p,o} = C(x_i^p, x_j), \quad 1 \le i \le n_p, 1 \le j \le n,$$

and $\Sigma^{p,p}$ is a $n_p \times n_p$ (unconditional) covariance matrix with entries

$$\Sigma_{ij}^{p,p} = C(x_i^p, x_j^p), \quad 1 \le i, j \le n_p.$$

The result in (3.7) can be derived by, first, noting that

$$\begin{pmatrix} z^p \\ y \end{pmatrix} \sim N\left(\begin{pmatrix} F^p\beta \\ F\beta \end{pmatrix}, \begin{pmatrix} \Sigma^{p,p} & \Sigma^{p,o} \\ (\Sigma^{p,o})^\top & \Sigma \end{pmatrix}\right),$$

and then using standard results of conditional distributions of multivariate Gaussian random variables. For the special case of univariate predictions at a single location x, the predictive distribution is given by

$$\begin{aligned} \pi(z(x)|y,\beta,\theta) &= N(\mu^p, \sigma^{2p}), \\ \mu^p &= f(x)^T\beta + c(x)^T\Sigma^{-1}\big(y - F\beta\big), \\ \sigma^{2p} &= C(x,x) - c(x)^T\Sigma^{-1}c(x), \end{aligned} \tag{3.8}$$

where

$$c(x) = \begin{pmatrix} C(x,x_1) \\ \vdots \\ C(x,x_n) \end{pmatrix}.$$

Figure 3.2 illustrates the posterior predictive distribution of a Gaussian processes for varying $n \in \{10, 20, 50, 100\}$.

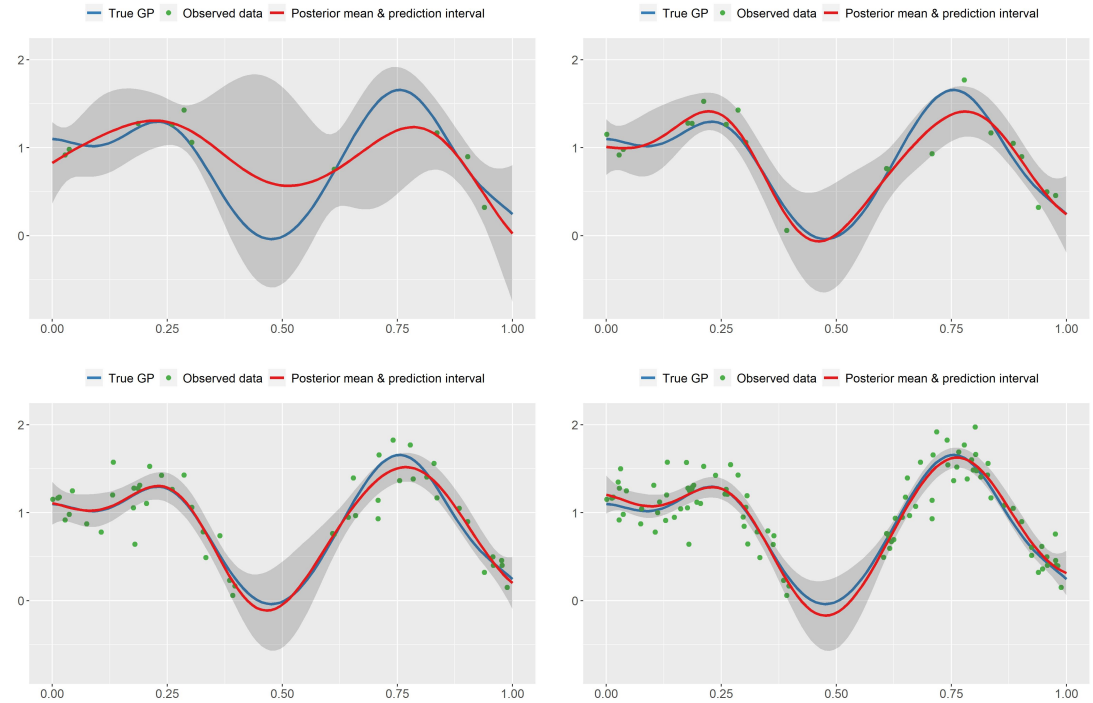


Figure 3.2: Illustration of the posterior predictive distribution of a Gaussian processes for varying $n \in \{10, 20, 50, 100\}$.

The predictive mean in (3.8) can also be derived as an optimal linear predictor without the Gaussian assumption. We look for the optimal predictor $\widehat{z}(x)$ minimizing the mean square error $\mathbb{E}\big((z(x) - \widehat{z}(x))^2\big)$ in the class of linear predictors which have the form

$$\begin{aligned} \widehat{z}(x) &= \lambda_0(x) + \sum_{i=1}^{n} \lambda_i(x)\big(y(x_i) - m(x_i)\big) \\ &= \lambda_0(x) + \lambda(x)^T(y - F\beta), \end{aligned}$$

where

$$\lambda(x) = \begin{pmatrix} \lambda_1(x) \\ \vdots \\ \lambda_n(x) \end{pmatrix}.$$

By the rules for the variance, we obtain

$$\begin{aligned} \mathbb{E}\big((z(x) - \widehat{z}(x))^2\big) &= \big(\mathbb{E}(z(x)) - \mathbb{E}(\widehat{z}(x))\big)^2 + \mathrm{Var}\big(z(x) - \widehat{z}(x)\big) \\ &= \big(m(x) - \lambda_0(x)\big)^2 + C(x,x) - 2\lambda(x)^T c(x) + \lambda(x)^T \Sigma \lambda(x). \end{aligned}$$

Thus, clearly $\lambda_0(x) = m(x) = f(x)^T\beta$. Setting the partial derivatives with respect to the other $\lambda(x)$ equal to zero leads to the equation system

$$\Sigma\lambda(x) = c(x).$$

This gives $\lambda(x) = \Sigma^{-1}c(x)$ which shows that $\widehat{z}(x)$ is equal to the predictive mean in (3.7) and (3.8). Furthermore,

$$\mathrm{Cov}\big(z(x_1^p) - \widehat{z}(x_1^p),\, z(x_2^p) - \widehat{z}(x_2^p)\big) = C(x_1^p, x_2^p) - c(x_1^p)^\top \Sigma^{-1} c(x_2^p)$$

for two prediction points x_1^p and x_2^p. Therefore, the covariance of the prediction errors $z(x_i^p) - \widehat{z}(x_i^p)$ at two locations x_1^p and x_2^p is the same as the conditional covariance in (3.7) under Gaussian assumptions. In particular, the variance of the prediction error $z(x) - \widehat{z}(x)$ equals the predictive variance obtained under Gaussian assumptions

$$\mathrm{Var}\big(z(x) - \widehat{z}(x)\big) = C(x,x) - c(x)^\top \Sigma^{-1} c(x).$$

Taking partial derivatives of $\mathbb{E}\big((z(x) - \widehat{z}(x))^2\big)$ with the respect to $\lambda_0(x)$ and $\lambda(x)$ and setting them to 0 shows that the optimal $\widehat{z}(x)$ is also characterized by the following orthogonality conditions:

$$\mathbb{E}\big(z(x) - \widehat{z}(x)\big) = 0, \qquad \mathbb{E}\big((z(x) - \widehat{z}(x))(y(x_i) - m(x_i))\big) = 0, \quad 1 \le i \le n. \tag{3.9}$$

This gives an alternative interpretation. Let $\mathcal{H} = L^2(\Omega, \mathcal{F}, \mathbb{P})$ denote the Hilbert space of square-integrable random variables with inner product

$$\langle U, V \rangle \;=\; \mathbb{E}(\,UV\,), \qquad \|U\|_2 \;=\; \big(\mathbb{E}[U^2]\big)^{1/2}.$$

The linear predictor $\widehat{z}(x)$ belongs to the finite-dimensional subspace

$$\mathcal{S} \;=\; \mathrm{span}\big\{\, 1,\; y(x_1) - m(x_1), \ldots, y(x_n) - m(x_n) \,\big\} \subset \mathcal{H}.$$

Minimizing the mean squared error $\mathbb{E}\big[(z(x) - \widehat{z}(x))^2\big]$ over $\mathcal{S}$ is, by definition, equivalent to orthogonally projecting $z(x)$ onto $\mathcal{S}$. By the projection theorem, the residual $e(x) := z(x) - \widehat{z}(x)$ is orthogonal to $\mathcal{S}$, i.e.

$$\langle e(x), 1 \rangle \;=\; 0, \qquad \langle e(x), y(x_i) - m(x_i) \rangle \;=\; 0, \quad i = 1, \ldots, n,$$

which are precisely the conditions in (3.9). In other words, $\widehat{z}(x)$ is the orthogonal projection of $z(x)$ onto $\mathcal{S}$.

In practice, β and θ are not known, and one can replace them by their empirical Bayes estimators $\hat{\beta}$ and $\hat{\theta}$ maximizing the marginal likelihood. However, uncertainty due to $\hat{\beta}$

and $\hat{\theta}$ is ignored when doing so. It turns out that uncertainty due to $\hat{\beta}$ can be easily taken into account. Recalling that

$$\widehat{z}(x) = f(x)^T\hat{\beta} + c(x)^T\Sigma^{-1}(y - F\hat{\beta}) \tag{3.10}$$

and plugging in the GLS estimator $\widehat{\beta}(\theta) = \big(F^\top\Sigma^{-1}F\big)^{-1}F^\top\Sigma^{-1}y$ from (3.6), we see that

$$\begin{aligned}\widehat{z}(x) &= \gamma(x)^\mathsf{T}y,\\ \gamma(x)^T &= \big(f(x)^T - c(x)^T\Sigma^{-1}F\big)\big(F^\mathsf{T}\Sigma^{-1}F\big)^{-1}F^T\Sigma^{-1} + c(x)^T\Sigma^{-1}.\end{aligned}$$

One can also show that this is again the best linear unbiased predictors (BLUP) as in the case when β was known.

Theorem 3.1 (BLUP / universal Kriging predictor)**.** *Let $y \in \mathbb{R}^n$ satisfy $\mathbb{E}(y) = F\beta$ and $\mathrm{Var}(y) = \Sigma$, with $\beta \in \mathbb{R}^p$ unknown and Σ known and positive definite. Let $z(x)$ be a stochastic process with*

$$\mathbb{E}\big(z(x)\big) = f(x)^\top\beta, \qquad \mathrm{Cov}\big(y, z(x)\big) = c(x), \qquad \mathrm{Var}\big(z(x)\big) = C(x,x).$$

Among all linear unbiased predictors $a^\top y$, the unique minimizer of $\mathbb{E}\big((z(x) - a^\top y)^2\big)$ is

$$\widehat{z}(x) \;=\; \gamma(x)^\top y, \qquad \gamma(x)^\top \;=\; \big(f(x)^\top - c(x)^\top\Sigma^{-1}F\big)\big(F^\top\Sigma^{-1}F\big)^{-1}F^\top\Sigma^{-1} \;+\; c(x)^\top\Sigma^{-1}.$$

Proof. First, $\widehat{z}(x)$ is clearly a linear homogeneous function of the observations y. Further,

$$\mathbb{E}\big(\gamma(x)^\mathsf{T}y\big) = \gamma(x)^TF\beta = f(x)^T\beta = \mathbb{E}(z(x)),$$

which shows that $\widehat{z}(x)$ is unbiased. In general, if $a^\mathsf{T}y$ is unbiased for all β, it must hold

$$\mathbb{E}[a^\mathsf{T}y] = a^\top F\beta \stackrel{!}{=} f(x)^\top\beta \quad\Longleftrightarrow\quad a^\top F = f(x)^\top.$$

Furthermore, $\mathbb{E}((z(x) - a^\top y)^2) = \mathrm{Var}(z(x) - a^\top y)$ and

$$\begin{aligned}\mathrm{Var}\big(z(x) - a^\top y\big) &= \mathrm{Var}(z(x)) + \mathrm{Var}(a^\top y) - 2\,\mathrm{Cov}(z(x), a^\top y)\\ &= C(x,x) + a^\top\Sigma a - 2a^\top c(x).\end{aligned}$$

Thus minimizing $\mathbb{E}((z(x) - a^\top y)^2)$ among all linear predictors is equivalent to minimizing

$$J(a) = a^\top\Sigma a - 2a^\top c(x) \quad \text{subject to } a^\top F = f(x)^\top.$$

We next introduce a Lagrange multiplier $\lambda \in \mathbb{R}^p$ and define

$$\mathcal{L}(a,\lambda) = a^\top\Sigma a - 2a^\top c(x) + 2\,\lambda^\top\big(F^\top a - f(x)\big).$$

Stationarity gives

$$\frac{\partial\mathcal{L}}{\partial a} = 2\Sigma a - 2c(x) + 2F\lambda = 0 \iff \Sigma a + F\lambda = c(x), \tag{3.11}$$

$$\frac{\partial\mathcal{L}}{\partial\lambda} = 2\big(F^\top a - f(x)\big) = 0 \iff F^\top a = f(x). \tag{3.12}$$

Due to (3.11), we have $a = \Sigma^{-1}\big(c(x) - F\lambda\big)$, and inserting this into (3.12) gives

$$F^\top\Sigma^{-1}\big(c(x) - F\lambda\big) = f(x) \iff \big(F^\top\Sigma^{-1}F\big)\lambda = F^\top\Sigma^{-1}c(x) - f(x).$$

Hence

$$\lambda = \big(F^\top \Sigma^{-1} F\big)^{-1}\big(F^\top \Sigma^{-1} c(x) - f(x)\big),$$

and

$$a(x) = \Sigma^{-1} c(x) \; + \; \Sigma^{-1} F\big(F^\top \Sigma^{-1} F\big)^{-1}\big(f(x) - F^\top \Sigma^{-1} c(x)\big).$$

Therefore the best linear unbiased predictors (BLUP) is

$$\widehat{z}(x) = a(x)^\top y = c(x)^\top \Sigma^{-1} y + \big(f(x)^\top - c(x)^\top \Sigma^{-1} F\big)\big(F^\top \Sigma^{-1} F\big)^{-1} F^\top \Sigma^{-1} y.$$

□

In addition, it follows that

$$\begin{aligned} \mathrm{Var}\big(z(x) - \widehat{z}(x)\big) =& C(x,x) - c(x)^\top \Sigma^{-1} c(x) \\ &+ \big(f(x) - F^T \Sigma^{-1} c(x)\big)^T \big(F^\mathsf{T} \Sigma^{-1} F\big)^{-1} \big(f(x) - F^T \Sigma^{-1} c(x)\big). \end{aligned} \tag{3.13}$$

The first two terms give the mean square prediction error in case β is known. The third term thus represents the additional uncertainty due to estimating β. Under Gaussian assumptions, $\pi(z(x)|y,\theta)$ is again Gaussian with mean and variance given in (3.10) and (3.13), respectively. This can be seen by first noting that

$$\pi(z(x)|y,\theta) = \int \pi(z(x)|y,\beta,\theta)\pi(\beta|y,\theta)d\beta,$$

where both $\pi(z(x)|y,\beta,\theta)$ and $\pi(\beta|y,\theta)$ are Gaussian when using a flat prior $\pi(\beta) \propto 1$ or, equivalently, a GLS estimator (3.6). This shows that $\pi(z(x)|y,\theta)$ is Gaussian with mean

$$\mathbb{E}(z(x)|y,\theta) = \mathbb{E}(\mathbb{E}(z(x)|y,\beta,\theta)|y,\theta) = f(x)^T\hat{\beta} + c(x)^T\Sigma^{-1}(y - F\hat{\beta})$$

and variance

$$\begin{aligned} \mathrm{Var}(z(x)|y,\theta) =& \mathbb{E}(\mathrm{Var}(z|y,\beta,\theta)|y,\theta) + \mathrm{Var}(\mathbb{E}(z|y,\beta,\theta)|y,\theta) \\ =& \mathbb{E}\big(C(x,x) - c(x)^\top \Sigma^{-1} c(x) \mid y,\theta\big) + \mathrm{Var}\big(\big(f(x)^T - c(x)^T\Sigma^{-1}F\big)\hat{\beta} \mid y,\theta\big) \\ =& C(x,x) - c(x)^\top \Sigma^{-1} c(x) \\ &+ \big(f(x) - F^T \Sigma^{-1} c(x)\big)^T \big(F^\mathsf{T} \Sigma^{-1} F\big)^{-1} \big(f(x) - F^T \Sigma^{-1} c(x)\big). \end{aligned}$$

In a fully Bayesian model, one uses prior distributions on the hyperparameters β and θ, and the posterior predictive distribution is

$$\pi(z^p|y) = \int \pi(z^p|y,\beta,\theta)\pi(\beta,\theta|y)d\beta d\theta.$$

The first term in the integrand on the right is Gaussian with mean and variance given in (3.7). In the fully Bayesian approach, one therefore averages the unknown parameters according to the posterior distribution instead of plugging in a point estimator. In particular, this takes uncertainty with respect to both unknown parameters β and θ into account in a coherent way. The drawback of this approach is that the integral with respect to the posterior cannot be computed in closed form. If one can simulate from the posterior, the integral can be approximated by an average and the prediction density $\pi(z^p|y)$ is approximated by a mixture of Gaussian densities. and they are again Gaussian.

3.4.4 Scalability to large data

For estimation and prediction, linear equation systems with the $n \times n$ matrix Σ need to be solved and the log-determinant of Σ needs to be calculated. These operations have computational complexity $O(n^3)$ and require $O(n^2)$ memory. I.e., this becomes computationally infeasible relatively quickly. There are multiple approximations that alleviate these computational burden. We will consider two types of approximation: Vecchia and low-rank inducing point approximations. For notational simplicity, we assume $m(x) = 0$ in the following. The extensions to general $m(x)$ are straightforward.

Vecchia approximations

In a Vecchia approximation, the Gaussian process prior density $\pi(y|\theta)$ is approximated as

$$\pi(y|\theta) = \prod_{i=1}^{n} \pi(y(x_i)|y(x_1), \ldots, y(x_{i-1}), \theta) \approx \prod_{i=1}^{n} \pi(y(x_i)|y_{N(i)}, \theta) = \tilde{\pi}(y|\theta),$$

where $y_{N(i)} \in \mathbb{R}^{|N(i)|}$ are subsets of $(y(x_1), \ldots, y(x_{i-1}))$, and $N(i) \subseteq \{1, \ldots, i-1\}$ with $|N(i)| \leq m$. For $i > m+1$, $N(i)$ is often chosen as the m nearest neighbors of x_i among $x_1, \ldots, x_{i-1}$. Note that

$$\pi(y(x_i)|y_{N(i)}, \theta) = \mathcal{N}(A_i y_{N(i)}, D_i),$$

where $A_i = \Sigma_{N(i),i}^T \Sigma_{N(i)}^{-1}$, $D_i = \Sigma_{i,i} - \Sigma_{N(i),i}^T \Sigma_{N(i)}^{-1} \Sigma_{N(i),i}$, $\Sigma_{N(i),i}$ is a sub-vector of $\Sigma = \mathrm{Cov}(y)$ with the i-th column and row indices $N(i)$, and $\Sigma_{N(i)}$ denotes sub-matrix of Σ consisting of rows and columns $N(i)$. Defining a sparse lower triangular matrix $B \in \mathbb{R}^{n \times n}$ with 1's on the diagonal, off-diagonal entries $B_{i,N_{(i)}} = -A_i$, and 0 otherwise, and a diagonal matrix $D \in \mathbb{R}^{n \times n}$ with D_i on the diagonal, one can show that the approximated density $\tilde{\pi}(y|\theta)$ is a Gaussian with a sparse precision matrix:

$$\tilde{\pi}(y|\theta) = \mathcal{N}(0, \tilde{\Sigma}),$$

where $\tilde{\Sigma}^{-1} = B^T D^{-1} B$ is sparse. Note that linear solves with $\tilde{\Sigma}$ can be done cheaply since

$$\tilde{\Sigma}^{-1} y = B^T D^{-1} B y,$$

and

$$\log\det\big(\tilde{\Sigma}\big) = -\log\det\big(\tilde{\Sigma}^{-1}\big) = -\log\det\big(D^{-1}\big) = \sum_{i=1}^{n} \log(D_i),$$

since B is a lower triangular matrix with 1's on the diagonal. Calculating a Vecchia approximation has $O(nm^3)$ computational and $O(nm)$ memory cost.

Vecchia approximations are sometimes called "local" approximations since the conditional distributions $\pi(y(x_i)|y(x_1), \ldots, y(x_{i-1}), \theta)$ are approximated by distributions that condition only on a set of neighbors. Often, very accurate approximations are obtained for small m's, e.g., $m = 20$, when p is small, say below 10 (recall $x \in D \subset \mathbb{R}^p$). For larger p's, Vecchia approximations are typically less accurate.

Inducing point approximations

Inducing point approximations approximate the covariance matrix Σ with a low-rank matrix. One chooses a set

$$\mathcal{S}^* = \{x_1^*, ..., x_m^*\}$$

of m distinct inducing points, or knots. Approaches for choosing the set of inducing points $\mathcal{S}^*$ include using k-means clustering, the cover-tree algorithm, or random subsampling from the data points $\{x_1, ..., x_n\}$. In the fully independent training conditional (FITC) approximation, also known as modified predictive process approximation in spatial statistics, one then approximates

$$\mathrm{Cov}(y) = \Sigma \approx \Sigma_{mn}^T \Sigma_m^{-1} \Sigma_{mn} + D_F,$$

where

$$D_F = \mathrm{diag}(\Sigma - \Sigma_{mn}^T \Sigma_m^{-1} \Sigma_{mn}),$$

$\Sigma_m = \left[C(x_i^*, x_j^*)\right]_{i=1:m, j=1:m} \in \mathbb{R}^{m \times m}$ is a covariance matrix of the inducing points, and $\Sigma_{mn} = \left[C(x_i^*, x_j)\right]_{i=1:m, j=1:n} \in \mathbb{R}^{m \times n}$ is a cross-covariance matrix between the inducing and data points. If the inducing points are chosen as a random subset of the data points and the diagonal correction D_F is dropped, the resulting approximation is also called the Nyström approximation.

Matrix inversion can then be done computationally efficiently using the Sherman-Woodbury-Morrison formula:

$$(\Sigma_{mn}^{\mathrm{T}} \Sigma_m^{-1} \Sigma_{mn} + D_F)^{-1} = D_F^{-1} - D_F^{-1} \Sigma_{mn}^{\mathrm{T}} (\Sigma_m + \Sigma_{mn} D_F^{-1} \Sigma_{mn}^{\mathrm{T}})^{-1} \Sigma_{mn} D_F^{-1},$$

and, by Sylvester's determinant theorem, the determinant can be calculated as

$$\det\left(\Sigma_{mn}^T \Sigma_m^{-1} \Sigma_{mn} + D_F\right) = \det\left(\Sigma_m + \Sigma_{mn} D_F^{-1} \Sigma_{mn}^{\mathrm{T}}\right) \cdot \det\left(\Sigma_m\right)^{-1} \cdot \det\left(D_F\right).$$

Such inducing point methods have $O(m^3 + nm^2)$ computational and $O(nm)$ memory cost. $m = 500$ or $m = 1000$ are commonly used choices.

In contrast to Vecchia approximations, inducing point methods are more typically accurate for larger p, i.e., high-dimensional inputs x. Local Vecchia and global inducing point approximations can also be combined to leverage the strength of both methods. For more details, see Gyger et al. (2025), https://arxiv.org/abs/2507.05064.

Chapter 4

Bayesian computation

Until now we have mainly worked with conjugate priors where the posterior is a standard distribution and moments or quantiles are either known explicitly or can be expressed by functions that can be coded efficiently in statistical software. But in the case of hierarchical models we already ended up with distributions that are not standard and contain integrals that cannot be computed analytically. Most applications today involve hierarchical models with many parameters, and often on the lowest level likelihoods occur for which conjugate priors do not exist. Thus the posterior is often high-dimensional with complex dependencies and does not belong to a standard family of distributions. In order to compute moments, quantiles, marginal densities, or posterior predictive distributions, we need approximate methods to compute integrals.

4.1 Laplace approximation

Laplace approximations are used for integrals of the form

$$\int h(\theta)q(\theta)d\theta$$

where q is a possibly unnormalized smooth density which is concentrated around its mode $\theta_0 = \arg\max \log q(\theta)$ and where h is an arbitrary smooth function. Because the gradient of $\log q$ at θ_0 is zero, we have

$$\log q(\theta) \approx \log q(\theta_0) - \frac{1}{2}(\theta - \theta_0)^T J(\theta_0)(\theta - \theta_0)$$

where J is minus the Hessian matrix with elements

$$J(\theta)_{ij} = -\frac{\partial^2}{\partial\theta_i \partial\theta_j} \log q(\theta).$$

Because $\log q$ is maximal at θ_0, $J(\theta_0)$ is positive definite. Expanding h into a first order Taylor series at θ_0, we obtain from the formula for the normalizing constant of a normal density

$$\begin{aligned}
\int h(\theta)q(\theta)d\theta &\approx h(\theta_0)q(\theta_0) \int \exp\left(-\frac{1}{2}(\theta - \theta_0)^T J(\theta_0)(\theta - \theta_0)\right) d\theta \\
&+ q(\theta_0)\frac{\partial h}{\partial\theta}(\theta_0)^T \int (\theta - \theta_0) \exp\left(-\frac{1}{2}(\theta - \theta_0)^T J(\theta_0)(\theta - \theta_0)\right) d\theta \\
&= h(\theta_0)q(\theta_0)(\det J(\theta_0))^{-1/2}(2\pi)^{p/2}. \qquad (4.1)
\end{aligned}$$

If we want to approximate a posterior expectation

$$\mathbb{E}(h(\theta) \mid x) = \frac{\int h(\theta)\pi(\theta)f(x \mid \theta)d\theta}{\int \pi(\theta)f(x \mid \theta)d\theta}. \tag{4.2}$$

we can use the Laplace approximation for the numerator and the denominator separately and take either $q(\theta) = f(x \mid \theta)$ or $q(\theta) = \pi(\theta)f(x \mid \theta)$. If h is strictly positive, we can also take $q(\theta) = h(\theta)\pi(\theta)f(x \mid \theta)$ in the numerator. All these variants lead to slightly different approximations. For an i.i.d. model with n samples, one can show that the approximation error of this approximation is of order $O(1/n)$ in both the numerator and the denominator of the ratio in (4.2). Further, if h is strictly positive and bounded away from 0 near θ_0, one can obtain an approximation error of $O(n^{-2})$ for the ratio in (4.2) when using the Laplace approximation for both the numerator and the denominator.

It is clear from the derivation that the Laplace approximation is good if q has a sharp peak at θ_0 and is much smaller everywhere outside of a neighborhood of θ_0. For a rigorous statement of a limit theorem justifying the use of the Laplace approximation, one has to consider a sequence of integrands that get sharper and sharper, that is $J(\theta_0) \to \infty$. This occurs for instance if we consider the posterior in an i.i.d. model with n observations where

$$q_n(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta),$$

θ_0 is the MLE $\widehat{\theta}_n$ and

$$J_n(\theta) = -\sum_{i=1}^{n} \frac{\partial^2}{\partial\theta\partial\theta^T} \log f(x_i \mid \theta)$$

is the observed Fisher information. By the law of large numbers

$$J_n(\theta) \approx nI(\theta).$$

Example 4.1. Bayes factors and the BIC.

Consider two different models M_1 and M_2 for i.i.d data $X_j \sim f_i(x \mid \theta_i)dx$ with parameters $\theta_i \in \mathbb{R}^{p_i}$ and priors π_i $(i = 1, 2)$. Then the Bayes factor of model 1 with respect to model 2 is

$$B_{12}(x_1, \ldots, x_n) = \frac{f(x_1, \ldots, x_n \mid M_1)}{f(x_1, \ldots, x_n \mid M_2)}, \quad f(x_1, \ldots, x_n \mid M_i) = \int \pi_i(\theta_i) \prod_{j=1}^{n} f_i(x_j \mid \theta_i)d\theta_i.$$

The Laplace expansion gives

$$f(x_1, \ldots, x_n \mid M_i) \approx \pi_i(\widehat{\theta}_i) \prod_{j=1}^{n} f_i(x_j \mid \widehat{\theta}_i)(\det(nI_i(\widehat{\theta}_i)))^{-1/2}(2\pi)^{p_i/2}$$

where $\widehat{\theta}_i$ is the MLE and $I_i(\theta)$ is the Fisher information in Model M_i. By the rules for the determinant, it follows that

$$\log f(x_1, \ldots, x_n \mid M_i) \approx \sum_{j=1}^{n} \log f_i(x_j \mid \widehat{\theta}_i) - \frac{p_i}{2} \log n + O(1).$$

The first term on the right measures how well the model fits the data and the second term is a penalty for model complexity. The two terms together can thus be used as a model selection criterion, the so-called the Bayesian information criterion (BIC). Two times the logarithm of the Bayes factor then approximately equals the difference between the BIC values of the two models.

Various extensions of the Laplace approximation are possible: If $q(\theta)$ is bimodal with modes at θ_0 and θ_1 that are well separated, we can use approximations around θ_0 and θ_1 separately. We can also use higher order Taylor approximations. To simplify the notation, assume that θ is one-dimensional. Then a fourth order Taylor approximation of $\log q$ together with $e^x \approx 1 + x$ gives

$$q(\theta) \approx q(\theta_0) \exp\left(-\frac{J(\theta_0)}{2}(\theta-\theta_0)^2\right)\left(1 + \frac{d^3 \log q(\theta_0)}{d\theta^3}\frac{(\theta-\theta_0)^3}{6} + \frac{d^4 \log q(\theta_0)}{d\theta^4}\frac{(\theta-\theta_0)^4}{24}\right).$$

The resulting integral can then be computed using results about higher moments of normal random variables.

4.2 Independent Monte Carlo methods

A Monte Carlo algorithm draws samples from a target distribution π on the basis of a sequence of i.i.d. uniform random variables (U^t). In applications, one uses pseudo-random numbers generated by a deterministic algorithm instead of truly random uniform sequences. How to get such pseudo-random numbers is a topic in itself, but we don't discuss this issue and just rely on the numbers provided by the software (e.g. by R). We assume that these pseudo-random numbers are good enough so that our results are not affected by this lack of true randomness.

Assume $(X^1, \dots X^N)$ is an i.i.d. sample from π. We use superscripts to identify the members of the sample because often π is a distribution on $\mathbb{R}^p$ for $p > 1$ and we use subscripts to indicate the different components of X. Also in Bayesian applications π is the posterior distribution, so the variable is θ and not X. But Monte Carlo methods are also used outside Bayesian statistics, and thus we use here the more familiar notation X for random variables. By the law of large numbers we can estimate the expectation μ_h of an arbitrary function $h(X)$ by the sample average

$$\mu_h = \int h(x)\pi(x)dx = \mathbb{E}_\pi(h(X)) \approx \bar{h}_N := \frac{1}{N}\sum_{t=1}^{N} h(X^t).$$

Moreover, by the central limit theorem

$$\frac{\sqrt{N}(\bar{h}_N - \mu_h)}{\sqrt{\sum_{t=1}^N (h(X^t) - \bar{h}_N)^2/N}} \overset{\text{approx}}{\sim} \mathcal{N}(0,1).$$

This can be used to compute an approximate (frequentist) confidence interval for μ_h. The convergence rate $N^{-1/2}$ is rather slow, but the advantage is that – in contrast to numerical approximations – it is independent of the dimension and doesn't require h to be smooth (only $\int h(x)^2\pi(x)dx < \infty$ is needed). These are essential advantages in Bayesian computations.

If π is one-dimensional and we can compute the quantile function

$$F_\pi^{-1}(u) = \inf\{x; F_\pi(x) \geq u\}, \quad F_\pi(x) = \int_{-\infty}^{x} \pi(x')dx',$$

then $X^t = F_\pi^{-1}(U^t) \sim \pi$. In higher dimensions we can use the method in principle sequentially to draw first X_1 from the first marginal, then X_2 from the conditional distribution of the second component given X_1 , etc. In Bayesian statistics this is rarely feasible because

the target π is known only up to a normalizing constant, and marginal or conditional quantiles cannot be computed explicitly.

There are two methods which both start by drawing a sample from some other distribution, the so-called proposal τ, and then convert it to a sample from the target π.

For rejection sampling, one assumes that the ratio of densities is bounded:

$$\frac{\pi(x)}{\tau(x)} \leq M < \infty.$$

Note that this implies that $supp(\pi) \subset supp(\tau)$ must hold. If we are given two independent i.i.d. sequences (Y^t) and (U^t) where $Y^t \sim \tau$ and $U^t \sim$ uniform(0,1), then the following algorithm produces an i.i.d. sequence (X^t) where $X^t \sim \pi$. To start, set $t = 1$ and $s = 1$ and repeat the following

- If $U^s \leq \frac{\pi(Y^s)}{M\tau(Y^s)}$ set $X^t = Y^s$, $t = t + 1$, $s = s + 1$ (i.e. accept Y^s);
- else set $s = s + 1$ whereas t is unchanged (i.e. reject Y^s).

To see that this algorithm is correct, we use the heuristic interpretation $\mathbb{P}(Y \in dx) = \tau(x)dx$ of the density τ. Then by Bayes' formula

$$\begin{aligned}\mathbb{P}(X \in dx) &= \mathbb{P}\left(Y \in dx \mid U \leq \frac{\pi(Y)}{M\tau(Y)}\right) \propto \mathbb{P}\left(U \leq \frac{\pi(Y)}{M\tau(Y)} \mid Y \in dx\right)\mathbb{P}(Y \in dx) \\ &= \frac{\pi(x)\tau(x)dx}{M\tau(x)} \propto \pi(x)dx.\end{aligned}$$

The algorithm works for any dimension, and τ and π need to be known only up to normalizing constants. One can show that the expected number of pairs that need to be sampled until the first Y is accepted equals M. In practice, the expected number of rejections is large unless τ is reasonably close to π, and in high dimensions it is usually difficult to find a proposal which is close to the the target and from which we can simulate.

The second method is called importance sampling. It corrects by weighting the sample values $Y^t \sim \tau$:

$$\int h(x)\pi(x)dx \approx \frac{\sum_{t=1}^N h(Y^t)w(Y^t)}{\sum_{t=1}^N w(Y^t)}, \quad w(x) \propto \frac{\pi(x)}{\tau(x)}.$$

By the law of large numbers applied to the denominator and the numerator it follows that almost surely

$$\frac{\sum_{t=1}^N h(Y^t)w(Y^t)}{\sum_{t=1}^N w(Y^t)} \to \frac{\int h(x)w(x)\tau(x)dx}{\int w(x)\tau(x)dx} = \frac{\text{const.}\int h(x)\pi(x)dx}{\text{const.}\int \pi(x)dx} = \int h(x)\pi(x)dx.$$

For the first equality, we need that $|h(x)|\pi(x) > 0$ implies $\tau(x) > 0$. One can also show that a central limit theorem holds where the limiting variance is

$$\frac{\int w(x)^2(h(x) - \int h(x')\pi(x')dx')^2\tau(x)d(x)}{(\int w(x)\tau(x)dx)^2}.$$

This asymptotic variance can easily be estimated from the sample. As with accept/reject, importance sampling can be used in any dimension and the proposal and the target need to be known only up to a normalizing constant. The weight function $w(x)$ does not need to be bounded, $\int w(x)^2\tau(x)dx < \infty$ is sufficient in order that the estimator has finite

variance for all bounded h. Still, if the normalized weights $w(Y^t)/\sum_s w(Y^s)$ are far from uniform, the above estimate becomes unreliable. We therefore need – like for accept/reject – a proposal that is not too far from the target.

If we want an unweighted sample instead of a weighted one, we can use resampling. We generate an additional sample (I^t) which takes values in $\{1, 2, \ldots, N\}$ with probabilities proportional to the weights $(w(Y^s))$:

$$\mathbb{P}(I^t = s) = \frac{w(Y^s)}{\sum_{r=1}^N w(Y^r)}$$

and set

$$Z^t = Y^{I^t}.$$

Thus values Y^t with large weights will likely be selected several times whereas those with small weights are likely to be not selected at all. This is easily seen to be correct because

$$\mathbb{E}\left(\sum_{t=1}^N h(Z^t) \mid Y^1, \ldots, Y^N\right) = \sum_{t=1}^N \sum_{s=1}^N \mathbb{P}(I^t = s) h(Y^s) = N \frac{\sum_{s=1}^N h(Y^s) w(Y^s)}{\sum_{r=1}^N w(Y^r)}.$$

Because the additional resampling step always increases the variance, direct use of it is not recommended. However, in Section 4.4.3, we will see a situation where we need an unweighted sample to proceed in a recursive algorithm.

4.3 Basics of Markov chain Monte Carlo

As mentioned in the previous section, drawing an i.i.d sample from a complicated distribution π is difficult, especially in high dimensions. Markov chain Monte Carlo generates a sequence of random variables (X^t) which are dependent and such that the distribution of X^t converges weakly to π as $t \to \infty$. Estimation of $\int h(x)\pi(x)dx$ is still based on a law of large numbers, but now for dependent random variables:

$$\int h(x)\pi(x)dx \approx \bar{h}_{N,r} = \frac{1}{N-r} \sum_{t=r+1}^N h(X^t).$$

Here r is a "burn-in" period which discards values X^t whose distribution is too far from the target π.

The random variables are constructed recursively: The initial value X^0 is arbitrary, and for each $t \geq 1$ X^t is a deterministic function of X^{t-1} and a uniform random variable U^t which is independent of $X^0, \ldots, X^{t-1}$

$$X^t = G(X^{t-1}, U^t).$$

Note that U^t is usually not one-dimensional but consists of several uniform variables. Because the dependence of X^t on previous random variables is only via X^{t-1}, the sequence (X^t) is called a Markov chain. The conditional distribution of X^t given X^{t-1} is called the transition kernel P of the chain

$$\mathbb{P}(X^t \in A \mid X^0, \ldots, X^{t-1}) = \mathbb{P}(X^t \in A \mid X^{t-1}) = P(X^{t-1}, A).$$

It is determined by the function G through

$$P(x, A) = \mathbb{P}(G(x, U) \in A) = \mathbb{P}(U \in \{u; G(x, u) \in A\}).$$

In particular, P does not depend on t because G is the same for all t. We therefore call the Markov chain time-homogeneous. In Markov process theory, one usually starts by specifying the transition kernel $P(x, A)$, the conditional probability that the next value of the chain is in A given that the current value is equal to x. It is always possible to construct a function G such that the above equation is satisfied. Because for Markov chain Monte Carlo, we need to draw from $P(x, .)$ for arbitrary values x, it is more natural to start with the concrete construction $X^t = G(X^{t-1}, U^t)$

In order to use Markov chain Monte Carlo to estimate expected values with respect to the target π, we need to find a transition kernel P such that we can draw from the distribution $P(x, .)$ for any x and such that for $N \to \infty$ the arithmetic mean of the $h(X^t)$ converges to $\int h(x)\pi(x)dx$. The general theory of Markov chains shows that the second requirement holds in a wide range of cases if the chain can reach all sets A with $\pi(A) > 0$ and if $X^{t-1} \sim \pi$ implies that $X^t \sim \pi$. If the second condition holds, we call π an invariant or stationary distribution for the transition kernel P. Because

$$\mathbb{P}(X^t \in A) = \mathbb{E}(\mathbb{P}(X^t \in A \mid X^{t-1})) = \mathbb{E}(P(X^{t-1}, A)),$$

π is stationary for P if

$$\pi(A) = \int \pi(x)P(x, A)dx \quad \forall A,$$

or, in the case where $P(x, .)$ has the density $p(x, y)$, if

$$\pi(y) = \int \pi(x)p(x, y)dx.$$

In the following, we introduce two basic recipes for constructing a transition kernel P which has a given target distribution π as stationary distribution. The first one is the so-called Gibbs sampler. For this we assume that $X \in \mathbb{R}^p$ and we divide X in k components $X = (X_1, X_2, \ldots, X_k)$. We denote the conditional density of the i-th component X_i of X given all the other components $X_{-i} = (X_j)_{j \neq i}$ by π_i:

$$\pi_i(x_i \mid x_{-i}) \propto \pi(x)$$

where $\propto$ means up to a term which does not contain x_i. This means that we can identify π_i by inspecting how the target density π depends on the i-th component. We don't need any integration. The densities π_i are also called "full conditionals" (because we condition on all other components). The Gibbs sampler depends on a "visiting schedule" $i_t \in \{1, 2, \ldots, k\}$ and iterates the following steps for $t = 1, 2, \ldots$

$$X^t_{i_t} \sim \pi_{i_t}(x_{i_t} \mid X^{t-1}_{-i_t}), \quad X^t_{-i_t} = X^{t-1}_{-i_t}.$$

In words, we leave all components of X^{t-1} unchanged except the one that is actually visited, and we update the visited component according to the conditional distribution of our target. By the definition of the conditional distribution, π is invariant for this transition kernel. The visiting schedule can be either deterministic or it can randomly select one of the components. In order that the chain can reach all sets, we have to visit each possible component infinitely often.

Example 4.2. I.i.d. normal observations. Assume that the X_i are i.i.d $\sim \mathcal{N}(\mu, \frac{1}{\tau})$ and that we use the prior

$$\pi(\mu, \tau) \propto \exp\left(-\frac{\kappa}{2}(\mu - \xi)^2\right) \tau^{\gamma-1} \exp(-\lambda\tau)$$

with hyperparameter $(\xi, \kappa, \gamma, \lambda)$. Because this is not a conjugate prior, the posterior

$$\pi(\mu, \tau \mid x_1, \ldots, x_n) \propto \tau^{n/2+\gamma-1} \exp\left(-\frac{\tau}{2}\sum(x_i-\mu)^2 - \lambda\tau - \frac{\kappa}{2}(\mu-\xi)^2\right)$$

is not a standard distribution. Because it is a two-dimensional distribution, we can get an idea about its shape by simply plotting the contours. The Gibbs sampler is also easy to use because

$$\begin{aligned}
\pi(\mu \mid \tau, x_1, \ldots, x_n) &\propto \exp\left(-\frac{\tau}{2}\sum_{i=1}^{n}(x_i-\mu)^2 - \frac{\kappa}{2}(\mu-\xi)^2\right) \\
&\propto \exp\left(-\frac{n\tau}{2}(\mu-\bar{x})^2 - \frac{\kappa}{2}(\mu-\xi)^2\right) \\
&\propto \exp\left(-\frac{n\tau+\kappa}{2}\left(\mu - \frac{n\tau}{n\tau+\kappa}\bar{x} - \frac{\kappa}{n\tau+\kappa}\xi\right)^2\right)
\end{aligned}$$

and

$$\pi(\tau \mid \mu, x_1, \ldots, x_n) \propto \tau^{n/2+\gamma-1} \exp\left(-\tau\left(\lambda + \frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2\right)\right).$$

Therefore the Gibbs sampler alternates between drawing μ from a normal distribution and τ from a Gamma distribution.

The Gibbs sampler requires that we can sample from the full conditionals. If this is not possible or it does not lead to a algorithm with satisfactory convergence properties, we can use the Metropolis-Hastings algorithm instead. It is based on the fact that any reversible distribution is also stationary. Here π is called reversible for the transition kernel P if

$$\int_A \pi(x)P(x,B)dx = \int_B \pi(x)P(x,A)dx \quad \forall A, B,$$

or in other words, if $X^t \sim \pi$, then

$$\mathbb{P}(X^t \in A, X^{t+1} \in B) = \mathbb{P}(X^{t+1} \in A, X^t \in B) \quad \forall A, B.$$

Choosing for B the whole space $\mathbb{R}^p$, it follows that a reversible distribution π is stationary. If $P(x, .)$ has the density $p(x, y)$ for any x, then reversibility is equivalent to

$$\pi(x)p(x,y) = \pi(y)p(y,x) \quad \forall x, y.$$

For any pair $x \neq y$, we can therefore choose one of the two values $p(x, y)$ and $p(y, x)$ arbitrarily, whereas the other one is determined by the reversibility equation. However, a solution obtained in this way does in general not satisfy $\int p(x, y)dy = 1$ for any x and thus is not the density of a transition kernel. To solve this problem, one can start with an arbitrary transition density q and then choose from the two possible solutions

$$p(x,y) = q(x,y),\ p(y,x) = \frac{\pi(x)q(x,y)}{\pi(y)}$$

and

$$p(x,y) = \frac{\pi(y)q(y,x)}{\pi(x)},\ p(y,x) = q(y,x)$$

the one which satisfies both $p(x, y) \leq q(x, y)$ and $p(y, x) \leq q(y, x)$ for any $x \neq y$. This solution can be written in the compact form

$$p(x,y) = q(x,y)a(x,y),$$

where

$$a(x,y) = \min\left(1, \frac{\pi(y)q(y,x)}{\pi(x)q(x,y)}\right).$$

It follows that $\int p(x,y)dy \leq \int q(x,y)dy = 1$ for any x, and one can put the missing mass on the diagonal, meaning that the chain does not move. Written in formulae, the transition kernel is

$$P(x,A) = \int_A p(x,y)dy + 1_A(x)\left(1 - \int p(x,y)dy\right).$$

Assuming that we can simulate from the transition density $q(x,.)$ for any x, the following algorithm generates a Markov chain with the transition kernel P:

- At time t generate $Y^t \sim q(X^{t-1},x)dx$ and $U^t \sim$ uniform(0,1), independently from each other and independently of previously generated variables.
- Set

$$X^t = \begin{cases} Y^t & \text{if} \quad U^t \leq a(X^{t-1},Y^t) \\ X^{t-1} & \text{else} \end{cases}$$

This is similar to rejection sampling, but the proposal depends on the most recent value, and in case of a rejection, we do not move. q is called the proposal distribution and a is called the acceptance probability.

The simplest choice of $q(x,.)$ is a normal density with mean x and an arbitrary covariance matrix Σ. In this case, $q(x,x') = q(x',x)$ so that the acceptance probability is simply $\min(1, \pi(x')/\pi(x))$. This means moving to value which is more likely than the current value is always accepted whereas the acceptance of a move to a less likely value is given by the likelihood ratio. The algorithm for a symmetric q is due to Metropolis et al. (1953) whereas the general case is due to Hastings (1970).

One can also combine Metropolis-Hastings with Gibbs by proposing a move in only one or a few components of x. The formula for accepting a move has the same structure as before.

The Gibbs and Metropolis algorithms are very flexible and can in principle handle most problems. However, judging how reliable the results are is not always easy. The situation differs from independent Monte Carlo in two aspects. First, because X^t is only approximately distributed according to the target π, there is a bias:

$$\mathbb{E}(\bar{h}_{N,r}) \neq \int h(x)\pi(x)dx.$$

Second, because successive values X^t are dependent, the variance is more complicated:

$$\mathrm{Var}\left(\bar{h}_{N,r}\right) = \frac{1}{(N-r)^2}\left(\sum_{t=r+1}^{N} \mathrm{Var}(h(X^t)) + 2\sum_{t=r+1}^{N}\sum_{s=1}^{N-t} \mathrm{Cov}(h(X^t), h(X^{t+s}))\right).$$

The pragmatic way to deal with these complications is to look at the times series plots of $h(X^t)$ or of components X_i^t versus t and to choose r such that the series "looks stationary" for $t \geq r$. Then one assumes that $X^t \sim \pi$ for $t \geq r$ so that there is no bias, $\mathrm{Var}(h(X^t))$ is independent of t and the covariances $\mathrm{Cov}(h(X^t), h(X^{t+s}))$ depend only on s and can be estimated by

$$\frac{1}{N-r}\sum_{t=r+1}^{N-s}(h(X^t) - \bar{h}_{N,r})(h(X^{t+s}) - \bar{h}_{N,r}).$$

The number of replicates N should then be large enough that these estimated covariances are close to zero for most lags s.

4.4 Some advanced computational methods

4.4.1 Adaptive MCMC

We have seen in the previous section that usually there are many transition kernels which have a given target π as the stationary distribution. The prime example is the Metropolis-Hastings algorithm with proposal $Y^t \sim \mathcal{N}(X^{t-1}, \Sigma)$ where Σ is an arbitrary positive definite covariance matrix. Although any choice of Σ leads to a consistent estimation of the quantities of interest $\int h(x)\pi(x)dx$, the choice of Σ has a large influence on the quality of the approximation for any finite number N of steps. In some prototype cases, it has been shown that if π is a p-dimensional distribution with covariance matrix $\mathrm{Cov}_\pi(X)$, then the "optimal" choice of Σ is given by

$$\Sigma = \frac{2.38^2}{p}\mathrm{Cov}_\pi(X).$$

A similar result says that the "optimal" choice is such that the average acceptance rate – after we have reached the stationary distribution – is 0.234, that is

$$\int \pi(x) \int q(x,y) \min\left(1, \frac{\pi(y)}{\pi(x)}\right) dydx = 0.234.$$

These criteria are considered as reasonable rules of thumb also in cases which differ from the ones in which they have been derived. The problem is however that they cannot be used directly because they depend on the unknown target π. A standard strategy is to have first an exploration phase where one tries out various values of Σ for an optimal acceptance rate or one estimates $\mathrm{Cov}_\pi(X)$ in order to obtain an estimate for Σ. In a second phase one then runs the algorithm with a fixed Σ determined from the experience gained in the exploration phase.

Adaptive MCMC combines the two phases by using a time dependent Σ^t which depends on the sequence of values $(X^0, X^1, \ldots, X^{t-1})$ generated so far. For instance, in order to approximate the first criterion mentioned above, one can take

$$\Sigma^t = \frac{2.38^2}{p}\frac{1}{t-1}\sum_{s=0}^{t-1}(X^s - \bar{X}^{t-1})(X^s - \bar{X}^{t-1})^T, \quad \bar{X}^{t-1} = \frac{1}{t}\sum_{s=0}^{t-1} X^s.$$

For the second criterion, let us assume that we only want to optimize the scale of Σ where the shape is fixed, e.g. $\Sigma = \sigma^2 I_p$. If we assume that the acceptance probability is a decreasing function of σ^2, then the following rule to choose $\sigma^{2,t}$ seems reasonable

$$\sigma^{2,t} = \begin{cases} r_t\sigma^{2,t-1} & \text{if} \quad \frac{1}{t-1}\sum_{s=0}^{t-2}\min(1, \frac{\pi(Y^{s+1})}{\pi(X^s)}) > 0.234, \\ \frac{1}{r_t}\sigma^{2,t-1} & \text{if} \quad \frac{1}{t-1}\sum_{s=0}^{t-2}\min(1, \frac{\pi(Y^{s+1})}{\pi(X^s)}) < 0.234. \end{cases}$$

Here Y^s is the proposed value in step s and $r_t \downarrow 1$.

The theoretical analysis of these and similar algorithms is delicate because one has to control the probability that this algorithm results in choosing a sequence Σ^t where some eigenvalues go to zero or to infinity. For more details, discussion and examples, see C. Andrieu, J. Thoms, A tutorial on adaptive MCMC, Stat. Comput. (2008) 18, 343 - 373.

4.4.2 Hamiltonian Monte Carlo

In some situations, the Metropolis-Hastings algorithm or the Gibbs sampler explore the target density π only slowly. This can be due to the fact that they either make only small steps or that proposals with big moves have a very low acceptance ratio. The idea of the Hamiltonian Monte Carlo (HMC) algorithm is to allow the simulated chain to make big moves that are still accepted with high probability.

In this section, we assume that $X \in \mathbb{R}^p$ and denote its density by π. Furthermore, we must be able to evaluate the gradient of $\log \pi$ efficiently.

HMC starts by considering a new target distribution $\tilde{\pi}$ on a space with doubled dimension

$$\tilde{\pi}(x,u) \propto \pi(x) \exp\left(-\frac{1}{2} u' M^{-1} u\right),$$

where M is a symmetric, positive-definite "mass" matrix. The matrix M is often diagonal or simply a scalar multiple of the identity matrix. In the following, we assume that $M = \mathrm{diag}(m_i)$.

The variables U_i are often called momentum variables. They can be thought of as auxiliary variables that allow the chain to make bigger moves. Clearly, if $(X,U) \sim \tilde{\pi}$, then $X \sim \pi$.

HMC is based on a deterministic, invertible map $G(x,u)$ that is volume preserving and keeps $\tilde{\pi}$ invariant

$$\left|\det \frac{\partial G(x,u)}{\partial x \partial u}\right| = 1, \quad \tilde{\pi}(G(x,u)) = \tilde{\pi}(x,u), \forall x,u.$$

The construction of such a map G is based on Hamiltonian mechanics which led to the name "Hamiltonian Monte Carlo". We define the Hamiltonian

$$H(x,u) = -\log \pi(x) + \sum_{i=1}^{p} \frac{u_i^2}{2m_i}.$$

In physical terms, x is the position, u is the momentum, $-\log \pi(x)$ is the potential and $\sum_{i=1}^{p} \frac{u_i^2}{2m_i}$ the kinetic energy. Note that the negative logarithm of the target is proportional to the Hamiltonian

$$-\log(\tilde{\pi}(x,u)) \propto H(x,u).$$

The transformation $G(x,u)$ is then the solution at time T of the ordinary differential equation

$$\begin{aligned} \frac{dx_i}{dt'} &= \frac{\partial H(x,u)}{\partial u_i} = \frac{u_i}{m_i} \\ \frac{du_i}{dt'} &= -\frac{\partial H(x,u)}{\partial x_i} = \frac{\partial \log \pi(x)}{\partial x_i}, \qquad 0 \le t' \le T, \end{aligned} \tag{4.3}$$

with initial condition (x,u). The value of T and the variances m_i are parameters of the method which need to be chosen by the user.

It is well known from basic mathematical physics (and straightforward to verify) that

$$\frac{d}{dt'} H(x(t'), u(t')) = 0, \quad \nabla \cdot \left(\frac{\partial H(x,u)}{\partial u_i}, -\frac{\partial H(x,u)}{\partial x_i}\right) = 0.$$

The first equation shows that G leaves $\tilde{\pi}$ invariant and the second that G is volume preserving, since the divergence of the vector field defined in (4.3) is zero. In addition, it can be shown that G is invertible.

For implementation, we need to solve the above differential equation by some discretization procedure. This unfortunately has the consequence that the transformation does not keep $\tilde{\pi}$ exactly invariant. The so-called "leapfrog method" however induces only small changes to $\tilde{\pi}$, it preserves volume exactly, and it is time-reversible (up to reflecting u), i.e., its implied mapping G is invertible. Thus it is possible to restore the exact invariance of $\tilde{\pi}$ by a Metropolis-Hastings step at the end with acceptance ratio

$$a((x,u),(x^*,u^*)) = \min\left(1, \exp\left(-H(x^*,u^*) + H(x,u)\right)\right), \tag{4.4}$$

where (x^*, u^*) is the newly proposed value and (x, u) the current one. Note that invertibility of G is needed for the reversibility of the implied transition kernel, and due to the volume preserving property, the acceptance ratio has the rather simple form above. In addition to T and the m_i's, the leapfrog method has another tuning parameter which is the step size ε of the discretization.

In order that the chain can sample from the whole space, the first step in a HMC iteration is to sample new values for the momentum variables. Note that if $(X, U) \sim \tilde{\pi}$ and $U' \sim N(0, \mathrm{diag}(m_i))$ is independent of (X, U), then also $(X, U') \sim \tilde{\pi}$. In other words, since in the first step, x is not changed and u is simulated from its correct conditional distribution, which is the same as its marginal distribution due to independence, this step also leaves $\tilde{\pi}$ invariant.

Algorithm 4.1 (Hamiltonian Monte Carlo algorithm)**.**
Choose $(X^0, U^0) = (x^0, u^0)$.
For $t = 1, 2, \ldots$

1a. Simulate $U' \sim N(0, \mathrm{diag}(m_i))$

1b. Use the leapfrog method (or any other method that results into an invertible G that is volume preserving) to generate a proposal $(X^*, U^*) = G(x^{t-1}, U')$

2. Simulate $V \sim uniform(0,1)$. If $V \leq a((x^{t-1}, U'), (X^*, U^*))$ set $X^t = X^*$, otherwise $X^t = x^{t-1}$.

Note that $a(\cdot, \cdot)$ is defined in Equation (4.4).

To better understand the strength of Hamiltonian Monte Carlo, let us assume that at some time t', we have $u_i > 0$, that is the i-th component of x moves to the right. If $\partial(log\pi(x))/\partial x_i > 0$ the i-th component moves into a region which is more likely under the target π, and by the Hamiltonian dynamics the momentum u_i increases and so we move faster. On the other hand, if $\partial(log\pi(x))/\partial x_i < 0$, we are going into a less likely region and the momentum u_i decreases and eventually gets reversed.

For more details, see e.g. Chapter 5 in Handbook of Markov Chain Monte Carlo, S. Brooks, A. Gelman, G.L. Jones and X.L. Meng (eds), Chapman and Hall, 2011.

4.4.3 Sequential Monte Carlo

This is a name used for methods which sample not from one target π, but from a sequence of related targets $\pi_0, \pi_1, \ldots, \pi_n = \pi$. The idea of sequential Monte Carlo (SMC) is essentially to apply importance sampling in an iterative way. We assume that at step

k, $X^{k,t}, t = 1, \ldots, N$, is a sample from π_k. This sample $(X^{k,t})$, is sequentially modified so that at the end we have a sample $(X^{n,t})$ from the original target π. As in the case of importance sampling, one has the possibility to generate weighted samples or equally weighted ones by applying a resampling step. We first focus on the later case.

Sequential Monte Carlo is in particular preferable over the Metropolis algorithm if π is multimodal since then sequential Monte Carlo has better chances to sample from regions around all modes.

Concerning the sequence (π_k), we can take, for instance, π_k as the posterior of θ given the first k observations, or we can take

$$\pi_k(x) \propto \pi(x)^{\phi_k}, \quad 0 \leq \phi_0 < \phi_1 < \ldots \phi_n = 1.$$

In this case, π_0 is close to a uniform distribution and we can for instance use rejection sampling to simulate from π_0.

In general, we assume that we can efficiently generate an initial sample $(X^{0,t})$ from π_0, e.g., using rejection or importance sampling. The sequential modification occurs by a propagation and an importance sampling step. Propagation means that at step k, we choose a transition density (kernel) p_k and generate

$$Y^{k,t} \sim p_k(X^{k-1,t}, y)dy, \quad \text{independently for } t = 1, 2, \ldots N.$$

Assuming that $(X^{k-1,t})$ is a sample from π_{k-1}, it follows that the distribution of $Y^{k,t}$ is given by

$$Y^{k,t} \sim \int \pi_{k-1}(x)p_k(x, y)dx \cdot dy.$$

In order to transform $(Y^{k,t})$ into a sample from π_k by importance sampling, we have to use the weights

$$w^{k,t} \propto \frac{\pi_k(Y^{k,t})}{\int \pi_{k-1}(x)p_k(x, Y^{k,t})dx}.$$

However, the integral in the denominator is in general not available analytically and thus the method cannot be used. The key idea is to look at the pairs $(X^{k-1,t}, Y^{k,t})$ which have $\pi_{k-1}(x)p_k(x, y)$ as density. We want to convert this distribution into one which has $\pi_k(y)$ as the second marginal. But all such densities have the form $\pi_k(y)q_{k-1}(y, x)$ where q_{k-1} is an arbitrary auxiliary backward transition kernel. Hence if we set

$$X^{k,t} = Y^{k,I^t}, \quad \mathbb{P}(I^t = s) \propto \frac{\pi_k(Y^{k,s})q_{k-1}(Y^{k,s}, X^{k-1,s})}{\pi_{k-1}(X^{k-1,s})p_k(X^{k-1,s}, Y^{k,s})},$$

$(X^{k,t})$ is a sample from π_k.

If one is willing to use weighted samples at all stages, we do not need a resampling step. We would then simply update the weights $w^{k,t}$

$$w^{k,t} \propto w^{k-1,t} \frac{\pi_k(X^{k,t})q_{k-1}(X^{k,t}, X^{k-1,t})}{\pi_{k-1}(X^{k-1,t})p_k(X^{k-1,t}, X^{k,t})}.$$

However, this sequential multiplication leads very quickly to unbalanced weights, and adding the resampling step gives more reliable approximations. Resampling helps to concentrate the computing effort in those region of the space where the densities π_k have their main mass.

For more details see, e.g., P. Del Moral, A. Doucet and A. Jasra, Sequential Monte Carlo Samplers, J. Royal Statist. Soc. B 68 (2006): 411-436.

4.4.4 Approximate Bayesian computation

For some models, evaluating the likelihood $f(x \mid \theta)$ is complicated or even impossible. In such cases, neither the Gibbs sampler nor Metropolis Hastings can be used to simulate from the posterior

$$\pi(\theta \mid x_{obs}) \propto \pi(\theta) f(x_{obs} \mid \theta).$$

But often, simulating a random variable $X \sim f(x \mid \theta)dx$ is much easier and we can therefore generate pairs $(\theta^t, X^t) \sim \pi(\theta) f(x \mid \theta) d\theta dx$. If X is discrete, we can use rejection sampling to simulate from the density proportional to

$$\pi(\theta) f(x \mid \theta) 1_{[x=x_{obs}]}$$

whose marginal is the posterior. I.e., we simply accept only pairs (θ^t, X^t) such that $X^t = x_{obs}$. Of course, it might take a very long time until we accept any pair at all, and in most cases X is continuous anyhow. But we can use the same idea if we replace the point mass at x_{obs} by a distribution which is concentrated near x_{obs}

$$\pi(\theta) f(x \mid \theta) \exp(-d(x, x_{obs})/\varepsilon)$$

where d is a metric on the space of observations. Instead of working with a fixed ε, one can also choose a sequence $\varepsilon_n \to 0$ with a rather large ε_0 and use a sequential Monte Carlo algorithm to produce samples of the corresponding targets.

For more details see e.g. M. Marin, P. Pudlo, C.P. Robert, and R.J. Ryder, Approximate Bayesian computational methods, Statistics and Computing, 22 (2012): 1167 - 1180.

Appendix A

Additional material

A.1 Combining quadratic forms

Let $x, a, b \in \mathbb{R}^n$, $A, B \in \mathbb{R}^{n\times n}$, A and B are symmetric, and $C = A+B$ is invertible. Then

$$(x-a)^T A(x-a) + (x-b)^T B(x-b) = (x-c)^T C(x-c) + (a-b)^T AC^{-1}B(a-b),$$

where $c = C^{-1}(Aa + Bb)$.

A.2 The change-of-variables formula

Assume that a random variable X has density $f_X(x)$. If g is an invertible and smooth transformation, then $Y = g(X)$ has the density

$$f_Y(y) = f_X(g^{-1}(y))|\det Dg^{-1}(y)|,$$

where Dg^{-1} denotes the Jacobian of g^{-1}.

A.3 Mean-square differentiability of stochastic processes

Definition A.1 (Mean-square derivative and m-times mean-square differentiable process)**.** *Let $z : \mathbb{R}^p \to L^2(\Omega, \mathcal{F}, \mathbb{P})$ be a (real-valued) stochastic process.*

(i) Mean-square derivative. *For a direction $v \in \mathbb{R}^p$, we say that z has a* mean-square derivative at $x \in \mathbb{R}^p$ in direction v *if there exists a random variable $D_v z(x) \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ such that*

$$\lim_{h\to 0} \mathbb{E}\left[\left(\frac{z(x+hv)-z(x)}{h} - D_v z(x)\right)^2\right] = 0.$$

For a multi-index $\alpha = (\alpha_1, \ldots, \alpha_p) \in \mathbb{N}_0^p$, the mean-square partial derivative $\partial^\alpha z(x)$ *is defined recursively by taking mean-square derivatives α_i times in direction e_i, provided the limits exist in L^2.*

(ii) m-times mean-square differentiable. *For $m \in \mathbb{N}_0$, we say that z is* m-times mean-square differentiable at x *if all mean-square partial derivatives $\partial^\alpha z(x)$ exist in L^2 for every multi-index α with $|\alpha| := \alpha_1 + \cdots + \alpha_p \le m$. If this holds for every $x \in \mathbb{R}^p$, we simply say that z is* m-times mean-square differentiable.